



GIS in Fire and Rescue:

Adopt a Hydrant and other public apps

FAIRFAX COUNTY FIRE AND RESCUE DEPARTMENT

Chip Galloway & John Woolverton

Introduction

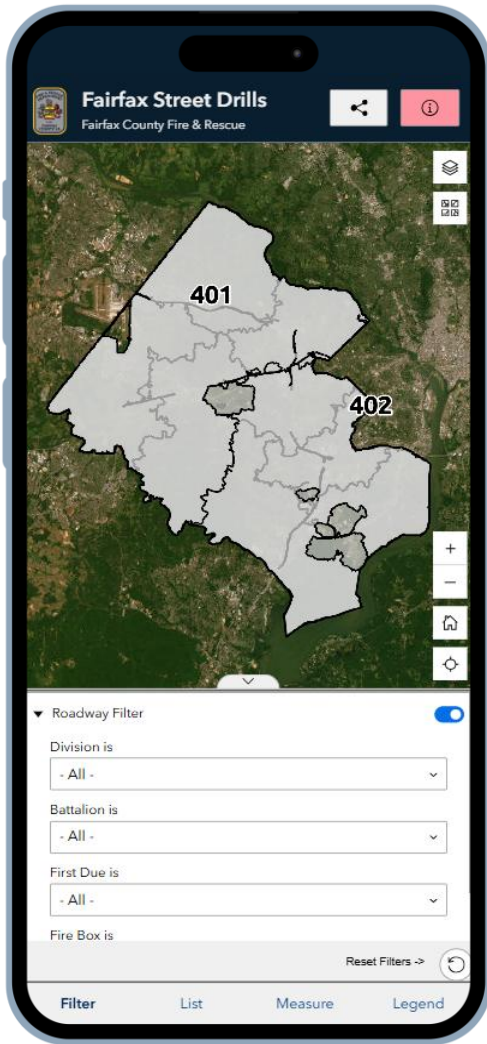
- Chip Galloway
- George Mason University BA in Geography 2008
- GIS Analyst since 2005
 - Federal contractor 2005 – 2006
 - Fairfax County Department of Transportation 2006 – 2008
 - Fairfax County Public Works (maintenance) 2008 – 2018
 - Fairfax County Public Works (planning) 2018 – 2021
 - Fairfax County Information Technology 2021 – 2024
 - Fairfax County Fire & Rescue Sept 2024



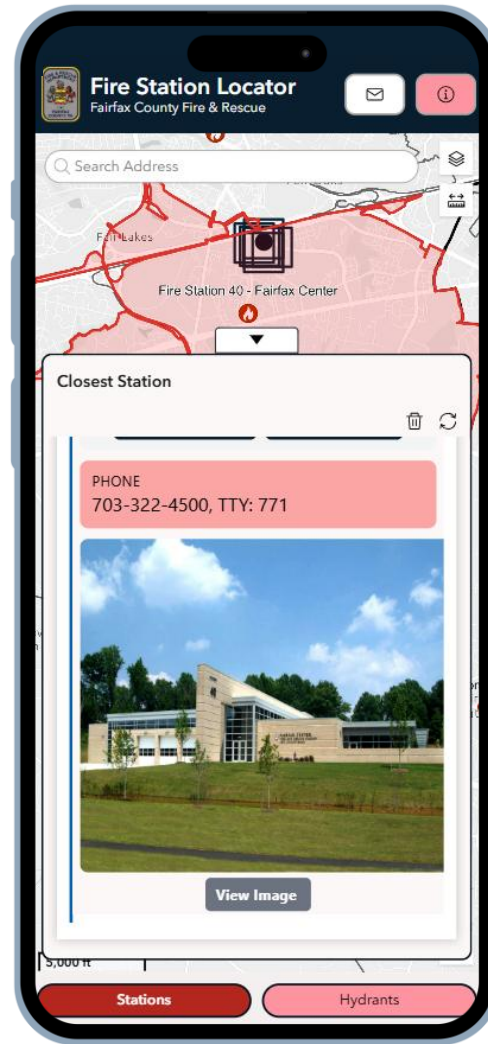
Introduction

- John Woolverton
- University of Mary Washington BA in Geography 2016
- GIS Analyst since 2017
 - Federal contractor 2017-2022 (Census Bureau – Geography Division)
 - Fairfax County Public Works (Directors Office - GIS) 2022 - 2025
 - Fairfax County Fire & Rescue April 2025

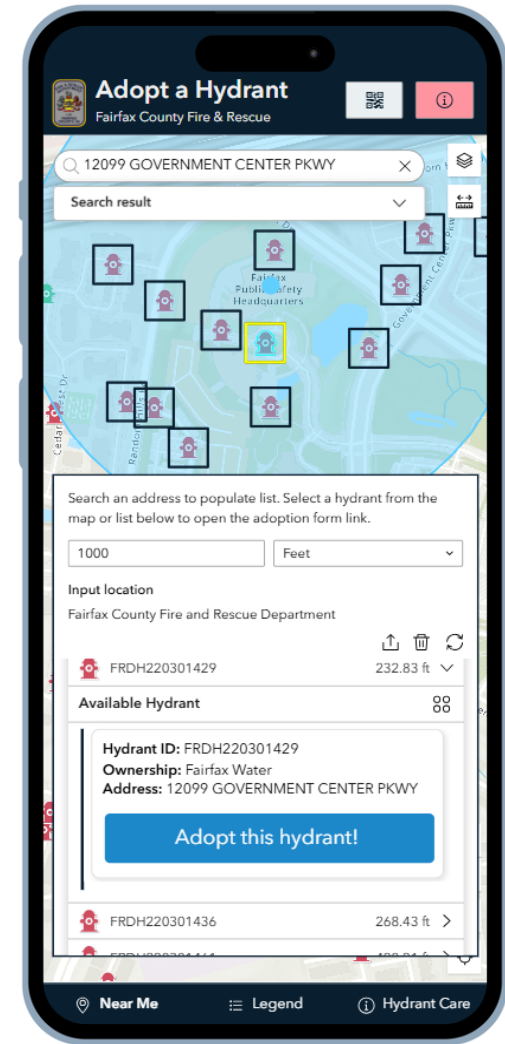




Street Drills



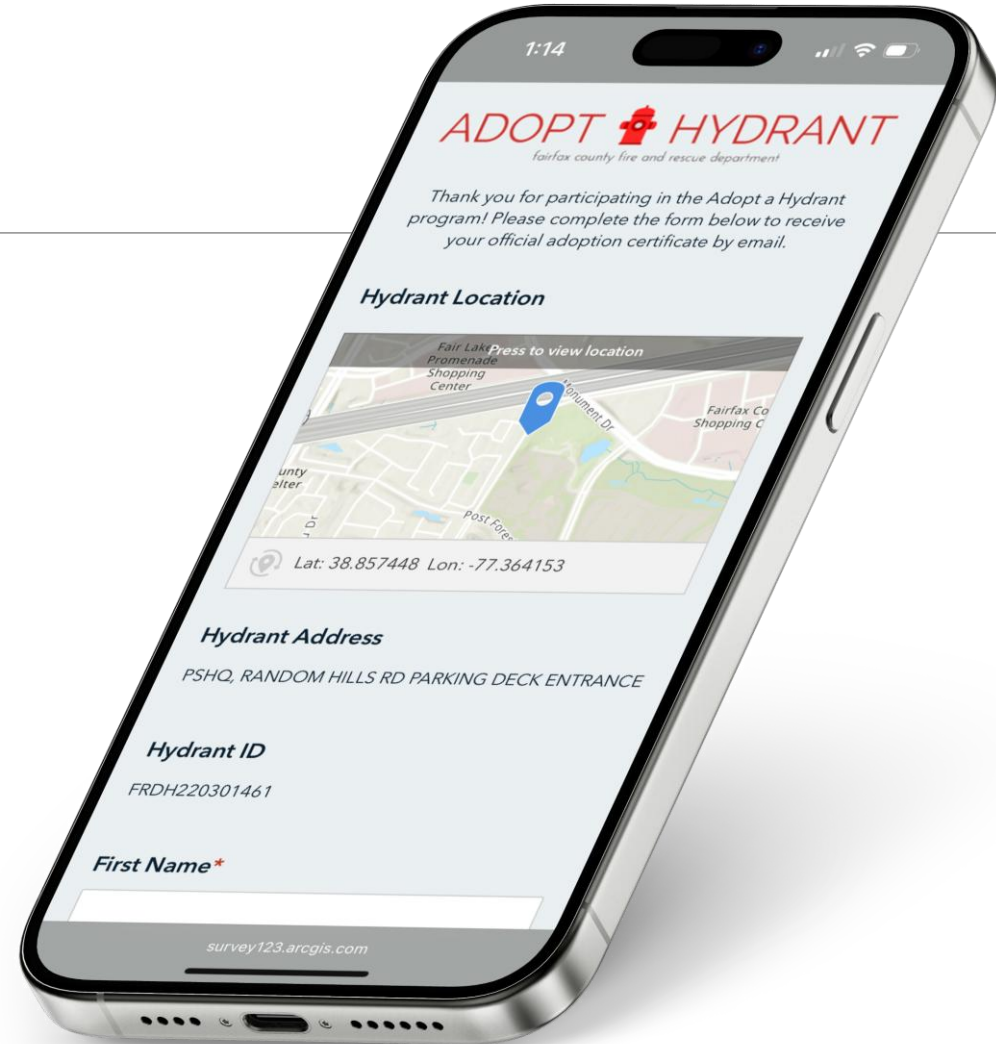
Fire Station Locator



Adopt a Hydrant

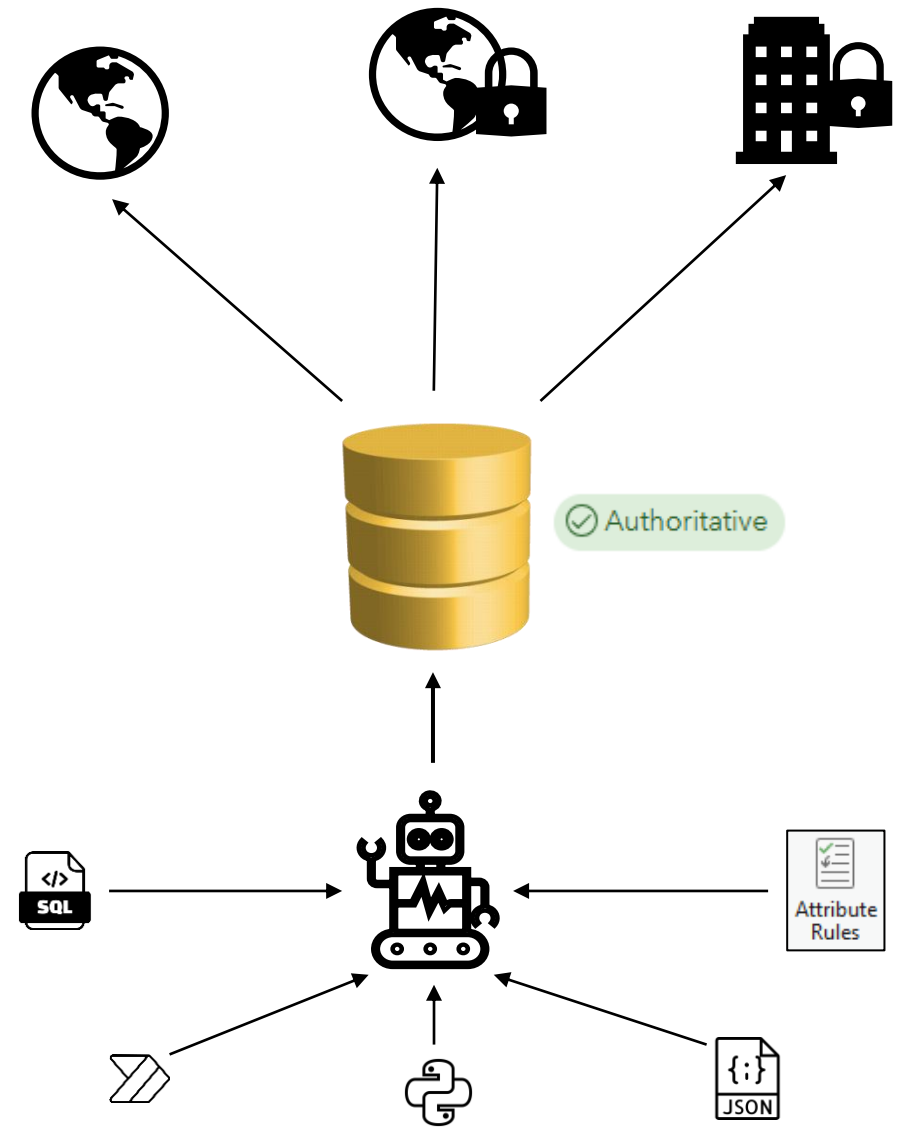
Public Facing Apps Overview

- Authoritative Data
- Applications
 - Street Drills
 - Station Locator
 - Adopt a Hydrant
- Desktop, Tablet, and Mobile Optimization



Authoritative Data

- External Data Sources
- Internal Data Sources
 - Comprehensive Schema
 - Centralized Storage
 - Documentation
 - Automating an Accurate Inventory
 - Publishing Process
 - Federated Services & Collaborations
 - Authoritative Services Supplying Multiple Apps



Authoritative Data External Data Sources

- Trusted datasets
- Reduces internal maintenance
- Near real-time updates
- Supports consistency

The screenshot displays the 'Fairfax County, Virginia - Open Geospatial Data' portal. The main heading is 'Roadways and Bridges'. Below this, there is a map preview showing a network of roads and bridges. To the left of the map is a vertical sidebar with the Fairfax County logo and the text 'Fairfax County Data Authoritative'. To the right of the map, there is a section titled 'Roadways and Bridges' with a green checkmark and the word 'Authoritative'. Below this, there is a section titled 'Fairfax County Authoritative Data' with the text 'County of Fairfax'. At the bottom of the map area, there are three buttons: 'View Map', 'Download', and 'More'. Below the map area, there is a 'Summary' section with two paragraphs of text. To the right of the summary is a 'Details' section with three items: 'Dataset' (Feature Layer), 'Unknown' (Info Updated: April 1, 2026 at 4:51:53 AM EDT), and 'April 1, 2026 at 4:52:01 AM EDT' (Data Updated). At the bottom of the details section, there is a 'January 1, 2013 at 12:00:00 AM EST' (Published Date).

Fairfax County, Virginia - Open Geospatial Data

Roadways and Bridges

✓ Authoritative

Fairfax County Authoritative Data
County of Fairfax

View Map Download More

Summary

The primary purpose of this data is to identify and depict the major transportation areas and assign attributes to them (paved and unpaved roads, bridges, and medians). Another purpose is to identify impervious surfaces.

Major transportation contains the primary road polygons and bridges in Fairfax County. The original data in this layer was captured during the 1997 data conversion effort for Fairfax County. After that an update capture was completed in 2014 using stereo models from the 2009 Virginia State imagery. Subsequent to that an update capture was completed in 2022 using stereo models from the 2017 Virginia

Details

Dataset
Feature Layer

Unknown
Info Updated: April 1, 2026 at 4:51:53 AM EDT

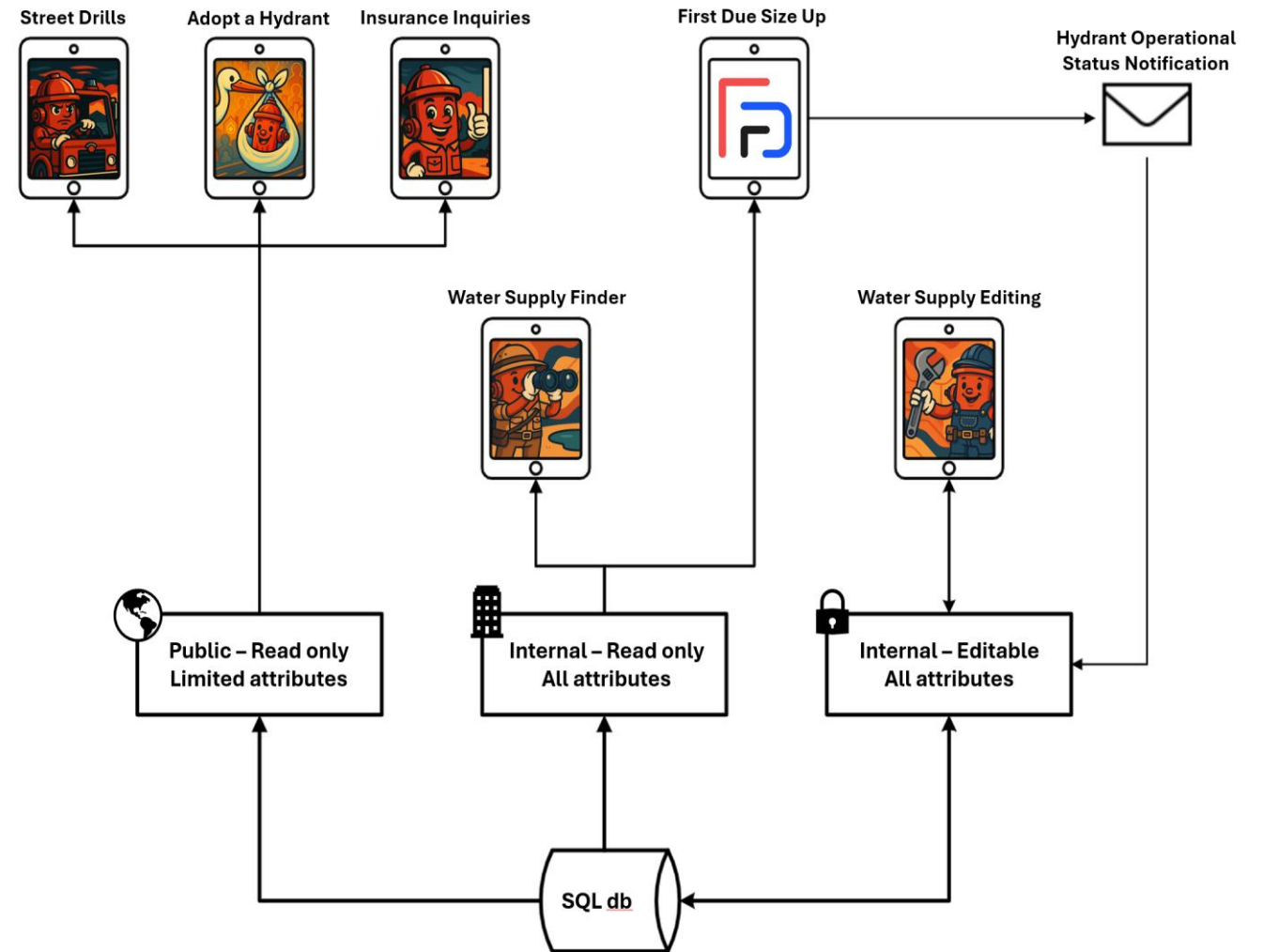
April 1, 2026 at 4:52:01 AM EDT
Data Updated

January 1, 2013 at 12:00:00 AM EST
Published Date

Authoritative Data

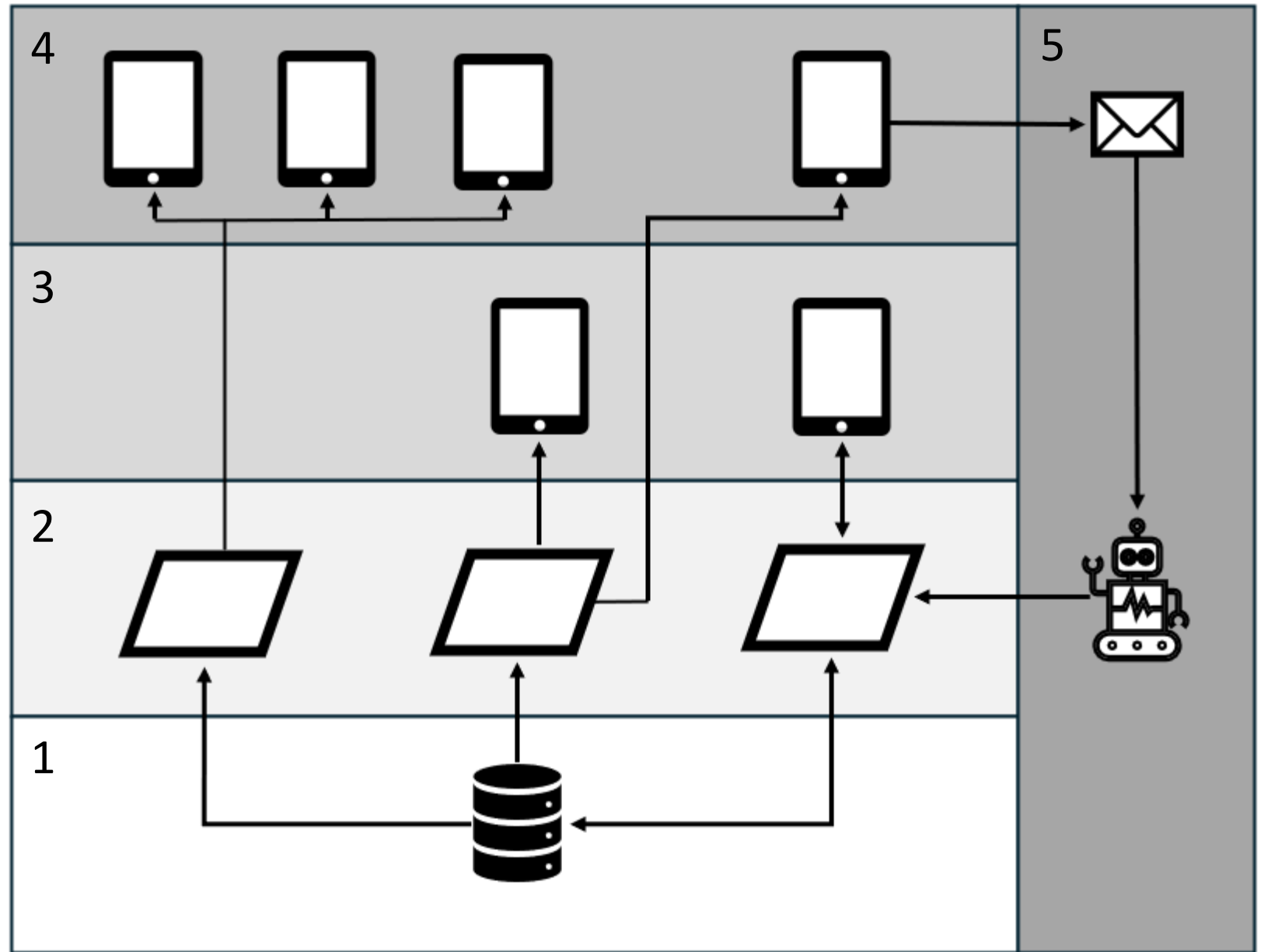
Internal Data

- Full control
- Tailored to operational needs
- Direct edit access
- Accountability
- Consistency across apps



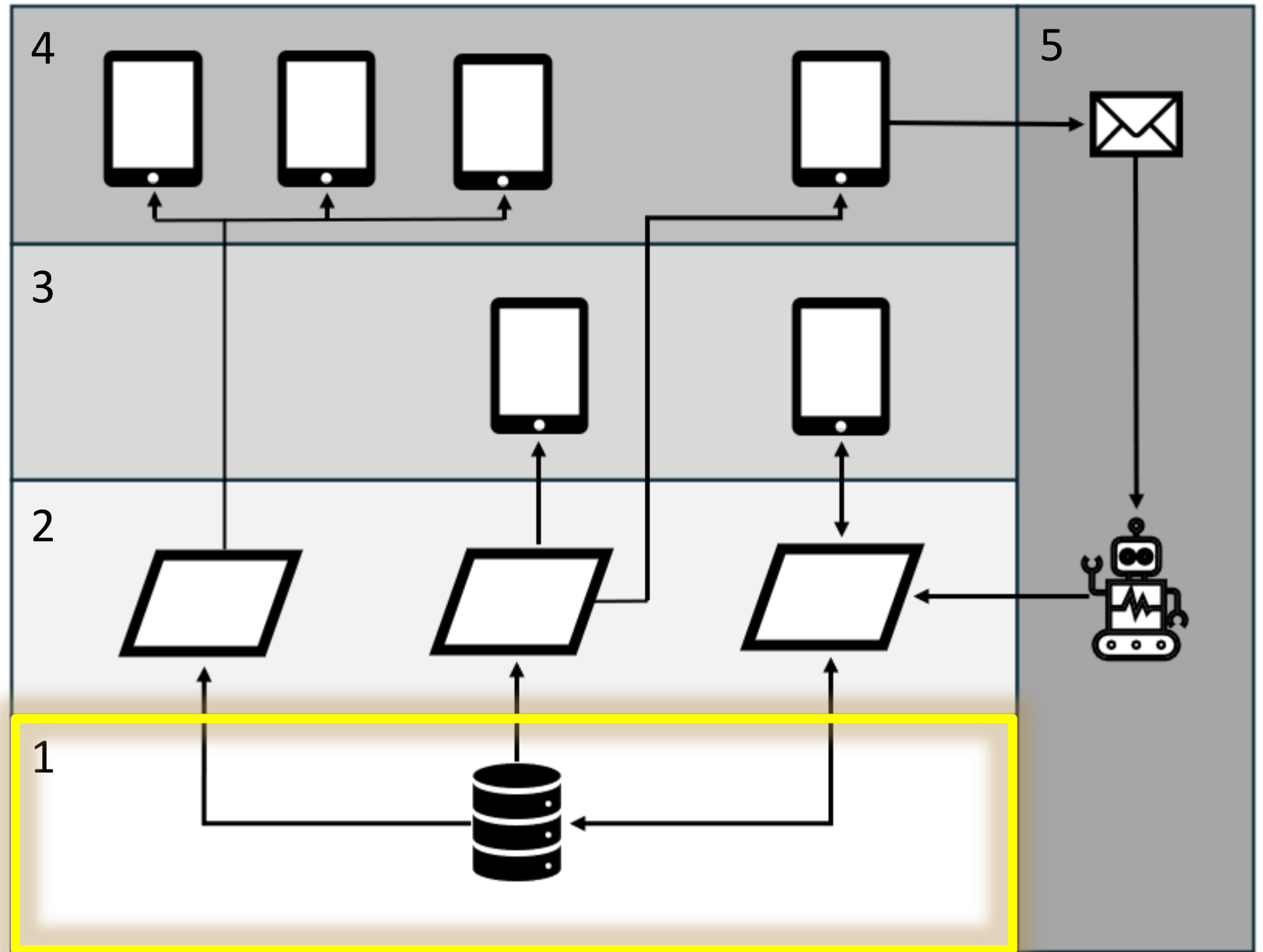
Water Supply Workflow

1. Database
2. Publishing
3. Enterprise Portal
4. ArcGIS Online
5. Automation



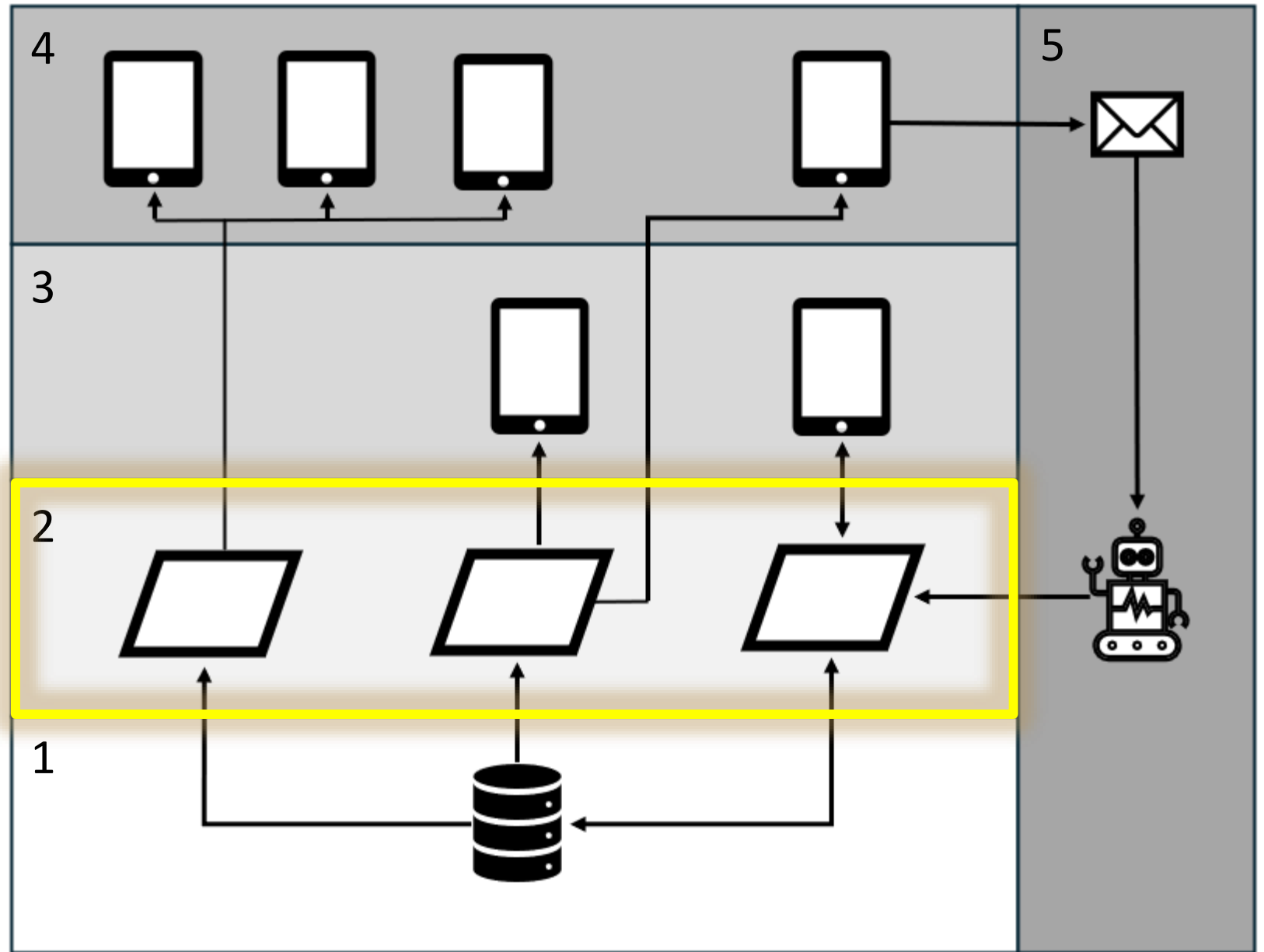
Water Supply Workflow

1. **Database**
2. Publishing
3. Enterprise Portal
4. ArcGIS Online
5. Automation



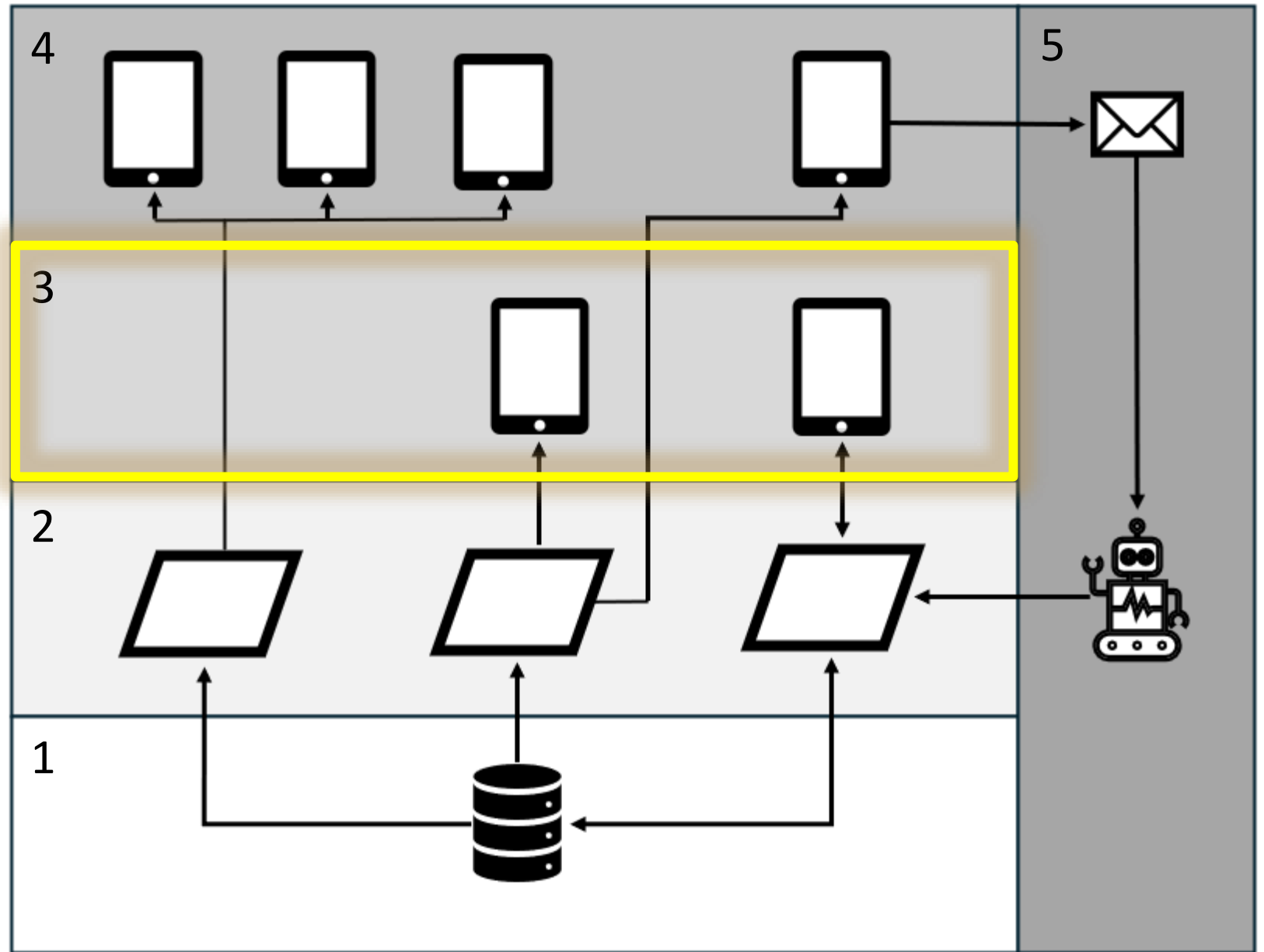
Water Supply Workflow

1. Database
2. **Publishing**
3. Enterprise Portal
4. ArcGIS Online
5. Automation



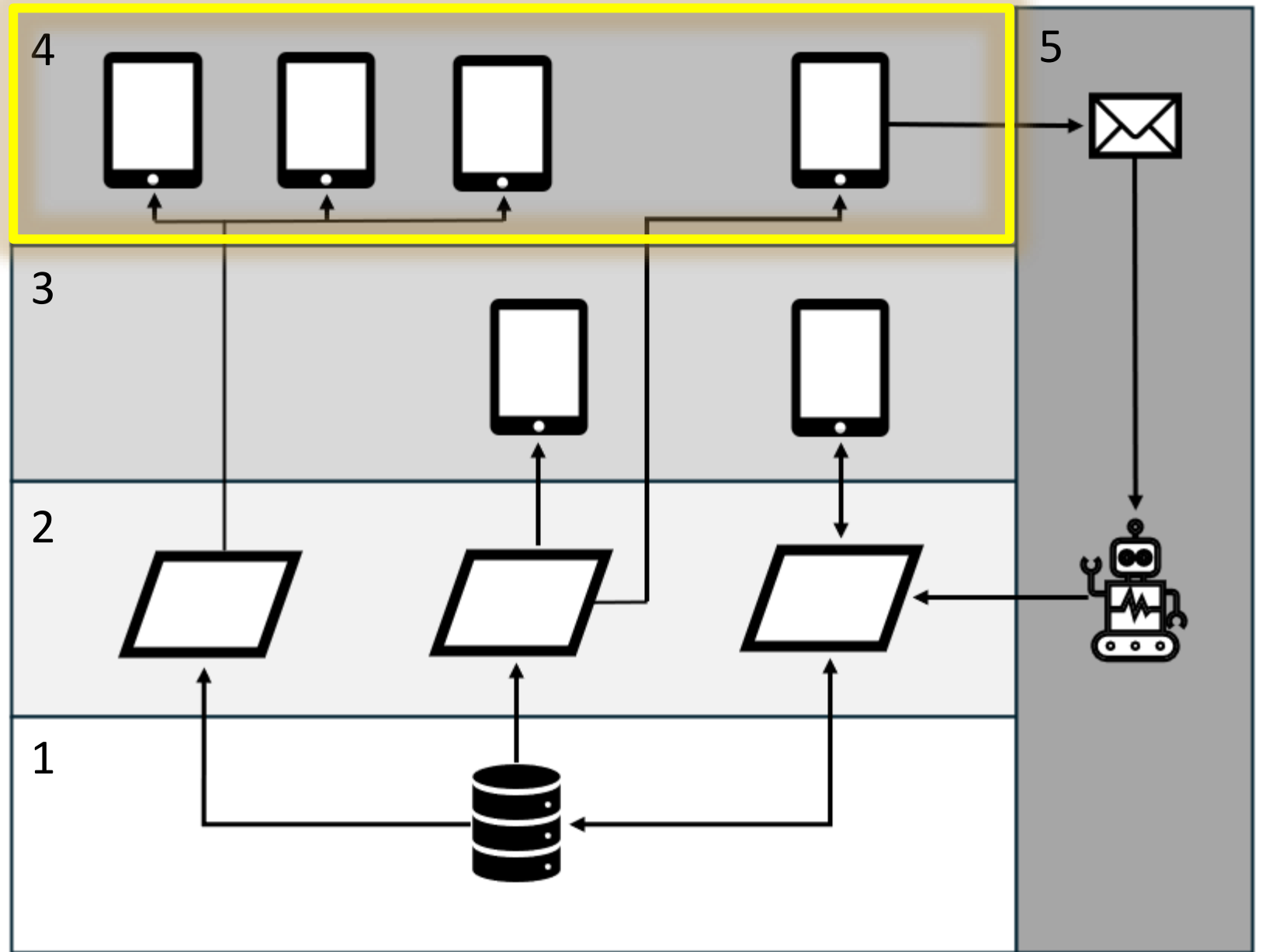
Water Supply Workflow

1. Database
2. Publishing
- 3. Enterprise Portal**
4. ArcGIS Online
5. Automation



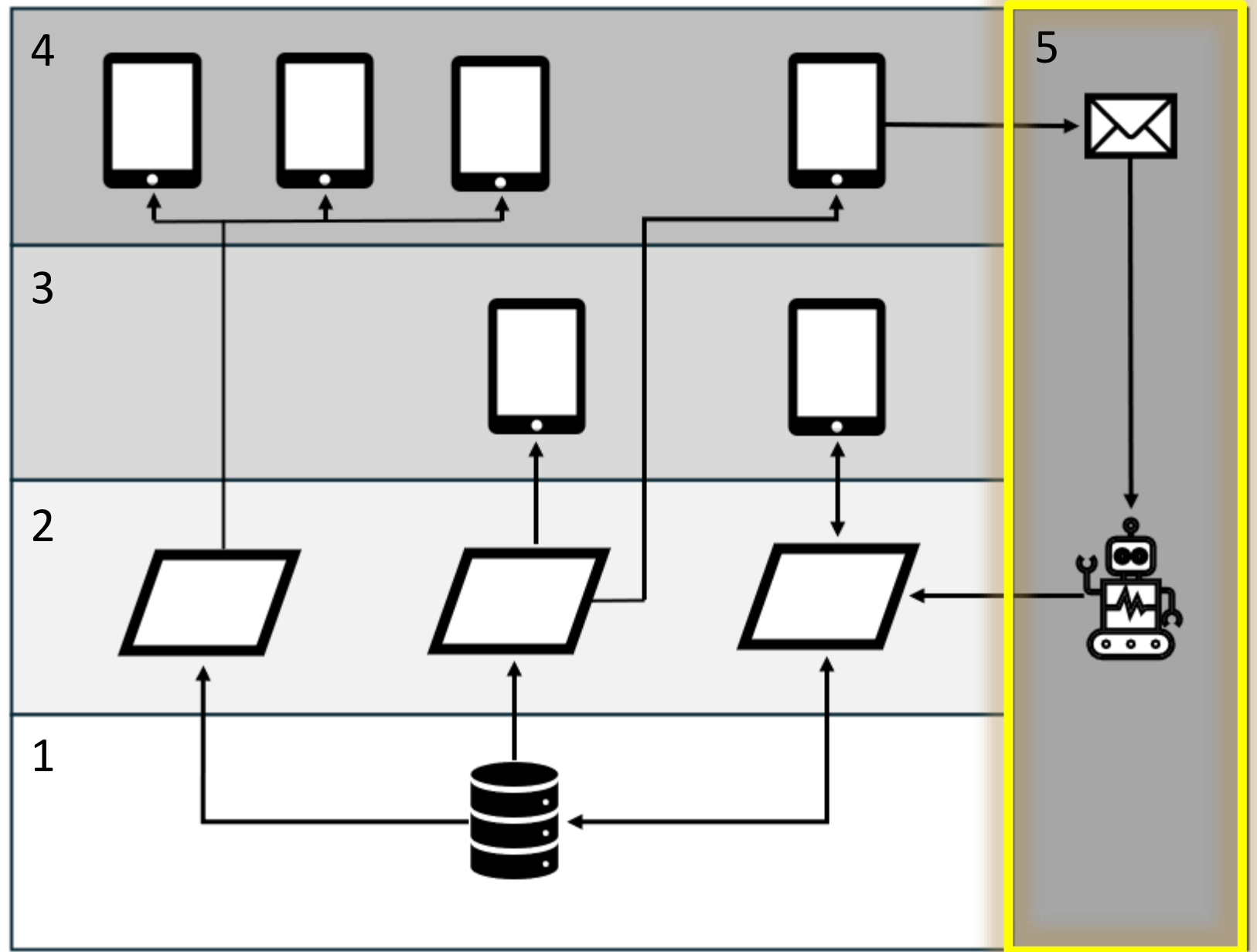
Water Supply Workflow

1. Database
2. Publishing
3. Enterprise Portal
4. **ArcGIS Online**
5. Automation



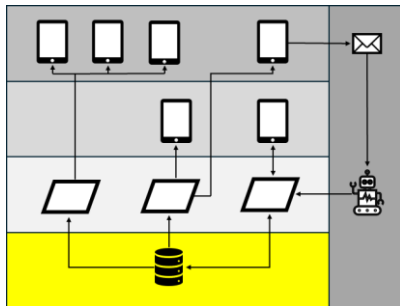
Water Supply Workflow

1. Database
2. Publishing
3. Enterprise Portal
4. ArcGIS Online
5. **Automation**



Water Supply Database

- Schema Development
- Centralized Storage
- Documentation
- Attribute rules
- Authoritative



Field Name	Alias	Data Type	Domain	Length
FRD_HYD_ID	FRD Hydrant ID	Text		20
HYDTYPE	Hydrant Type	Text	FRD_HYD_TYPE	15
HYD_SUBTYPE	Water Supply Subtype	Text	FRD_HYD_SUBTYPE	15
STATUS	Status	Text	FRD_HYD_STATUS	20
FLOW	Modeled Flow Rate	Long		
FIRST_DUE	First Due	Short		
FIRE_BOX	Fire Box	Text		50
ADDRESS	Nearest Address	Text		80
LOC_DESC	Location Description	Text		255
LOC_DETAILS	Location Details	Text		50
ZIP	ZIP Code	Text		10
JURISDICTION	Jurisdiction	Text		25
CAD_CODE	CAD Code	Text		30
IN_FDSU	In First Due Size Up	Text	YES_NO	3
DATA_SOURCE	Data Source	Text		20
SOURCE_ID	ID From Data Source	Text		30
SOURCE_TYPE	Source Type	Text		50
SOURCE_STATUS	Source Hydrant Status	Text		20
MAINTAINED_BY	Maintained By	Text		30
OWNERSHIP	Ownership	Text	FRD_HYD_OWNER	50
DATA_STEWARD	Data Steward	Text		20
X_COORD	X State Plane	Double		
Y_COORD	Y State Plane	Double		
X_DD	X Decimal Degrees	Double		
Y_DD	Y Decimal Degrees	Double		
WHAT_THREE_WORDS	What 3 Words	Text		40
VERIFIED_METHOD	Verified Method	Text		10
VERIFIED_BY	Verified By	Text		3
VERIFIED_DATE	Last Verified Date	Date		
DATA_NOTES	Data Notes	Text		255
HYDRANT_NOTES	Hydrant Notes	Text		255
BUILDING_QUADRANT	Building Quadrant	Text	FRD_HYD_QUAD	1
SEQ_NUM	Sequential Number	Long		

Refined schema



SQL Database

Metadata

WATER SUPPLY

Thumbnail

Tags

FFX, Fairfax County, Virginia, VA, Fire and Rescue Department, Data Analytics Strategy Management Division, FCFRD, FFXCO, FRD, Fire, Public Safety, Fire and Rescue, Water Supply, Hydrant

Summary

This layer contains water supply sources used for firefighting. Data supports Fairfax County Computer Aided Dispatch (CAD), First Due Size Up (FDSU), Insurance Inquiries, Adopt a Hydrant, Street Drills, Water Supply Finder, and Hydrant Verification applications.

Description

The layer is a combination of multiple data sources from internal jurisdictions including but not limited to the towns of Vienna, Clifton, and Herndon, and Ft. Belvoir. It also includes some private hydrants from neighboring jurisdictions which include the City of Fairfax and Falls Church, Loudoun County, Prince Williams County, Montgomery County (MD), Prince George County (MD), Washington, D.C. Most of this data is provided by the Fairfax Water Authority.

Contact: Fire and Rescue Department

Data Accessibility: Publicly Available

Update Frequency: As Needed

Last Revision Date: 3/10/2025

Creation Date: 1/1/2009

Feature Dataset Name: FIREMGR.FIRE_RESCUE_PUBLIC

Layer Name: FIREMGR.WATERSUPPLY

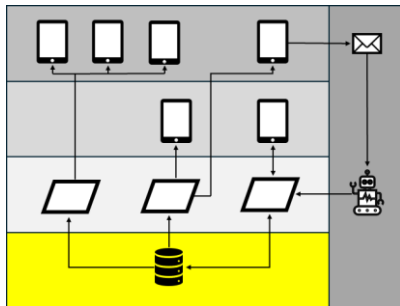
Credits

Fairfax County, Virginia; Fire and Rescue Department;

Metadata & Data dictionary

Water Supply Database

- Local to SQL
- Schema Development
- Documentation
- Attribute rules**
- Authoritative



Field Name	Alias	Data Type	Domain	Length
FRD_HYD_ID	FRD Hydrant ID	Text		20
HYDTYPE	Hydrant Type	Text	FRD_HYD_TYPE	15
HYD_SUBTYPE	Water Supply Subtype	Text	FRD_HYD_SUBTYPE	15
STATUS	Status	Text	FRD_HYD_STATUS	20
FLOW	Modeled Flow Rate	Long		
FIRST_DUE	First Due	Short		
FIRE_BOX	Fire Box	Text		50
ADDRESS	Nearest Address	Text		80
LOC_DESC	Location Description	Text		255
LOC_DETAILS	Location Details	Text		50
ZIP	ZIP Code	Text		10
JURISDICTION	Jurisdiction	Text		25
CAD_CODE	CAD Code	Text		30
IN_FDSU	In First Due Size Up	Text	YES_NO	3
DATA_SOURCE	Data Source	Text		20
SOURCE_ID	ID From Data Source	Text		30
SOURCE_TYPE	Source Type	Text		50
SOURCE_STATUS	Source Hydrant Status	Text		20
MAINTAINED_BY	Maintained By	Text		30
OWNERSHIP	Ownership	Text	FRD_HYD_OWNER	50
DATA_STEWARD	Data Steward	Text		20
X_COORD	X State Plane	Double		
Y_COORD	Y State Plane	Double		
X_DD	X Decimal Degrees	Double		
Y_DD	Y Decimal Degrees	Double		
WHAT_THREE_WORDS	What 3 Words	Text		40
VERIFIED_METHOD	Verified Method	Text		10
VERIFIED_BY	Verified By	Text		3
VERIFIED_DATE	Last Verified Date	Date		
DATA_NOTES	Data Notes	Text		255
HYDRANT_NOTES	Hydrant Notes	Text		255
BUILDING_QUADRANT	Building Quadrant	Text	FRD_HYD_QUAD	1
SEQ_NUM	Sequential Number	Long		

Schema

Fire Box

Calculation

2.3

Rule Name

Description

Subtype

Field

☒ Editable

Expression ☒

```
// Access the Fire_Box layer and retrieve the COG_BOX_NUM field
var gridLayer = FeatureSetByName($datastore, "Fire_Box", ["COG_BOX_NUM"], true);

// Find the first feature that intersects with the current feature
var matchingFeature = First(Intersects(gridLayer, $feature));

// Check if a matching feature is found
if (!IsEmpty(matchingFeature)) {
    var cogBoxNum = matchingFeature["COG_BOX_NUM"];

    // Ensure the number is 6 digits, padding with leading zeros if necessary
    return Text(cogBoxNum, "000000");
}

// Return a default value (e.g., "000000") if no match is found
return "000000";
```

Triggers

Insert ☒

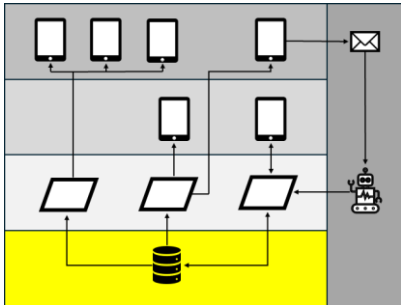
Update ☒

Delete ☐

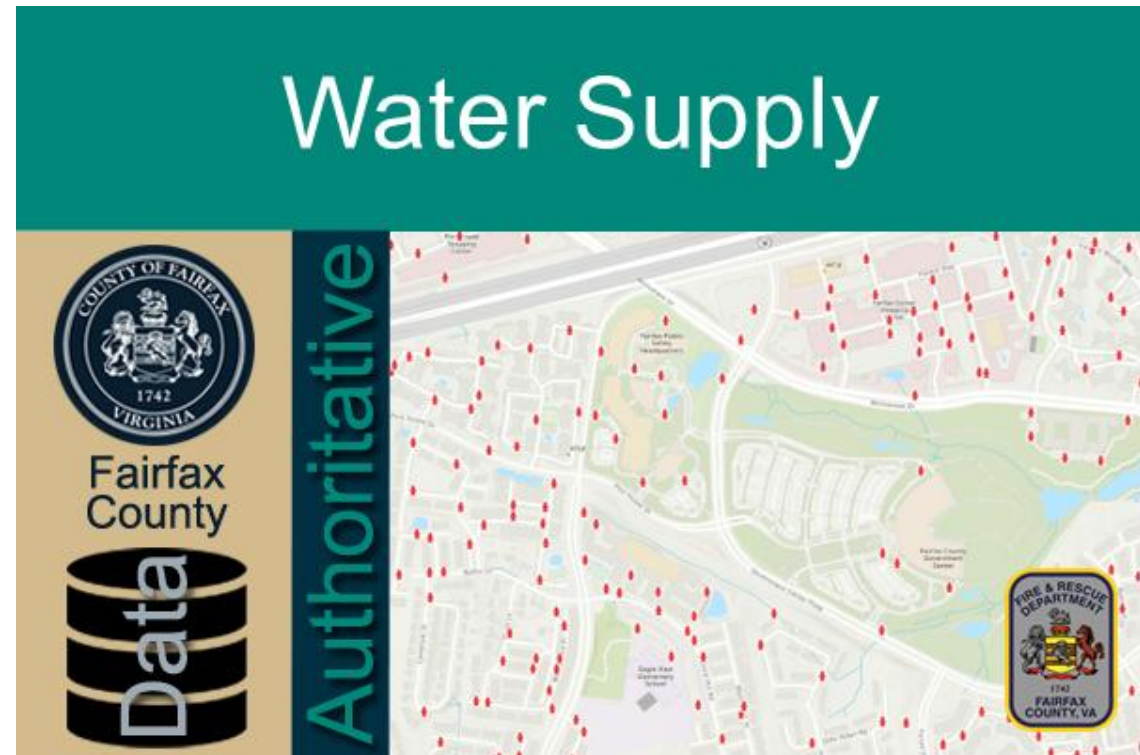
Attribute Rules

Water Supply Database

- Local to SQL
- Schema Development
- Documentation
- Attribute rules
- **Authoritative**



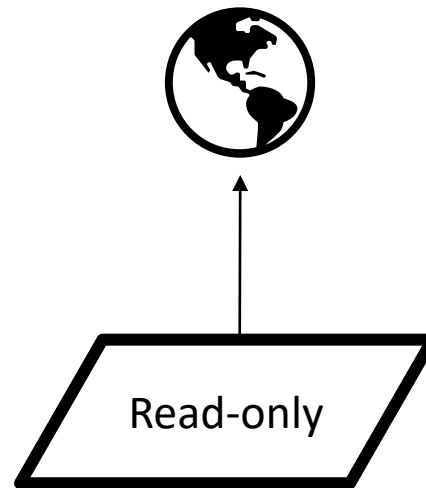
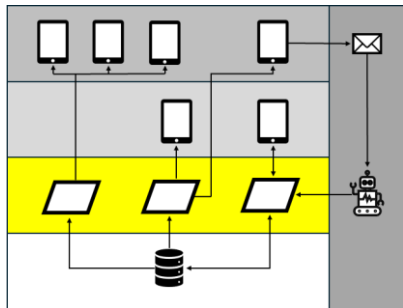
✓ Authoritative



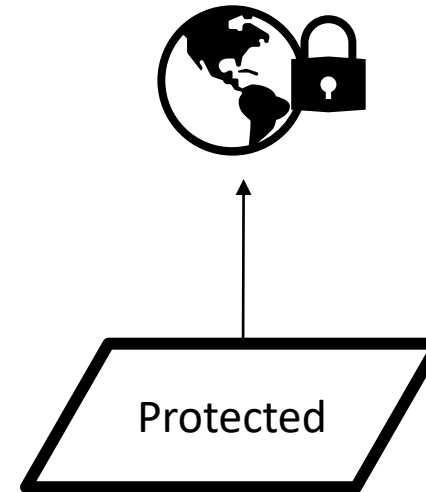
Authoritative thumbnail

Water Supply Publishing

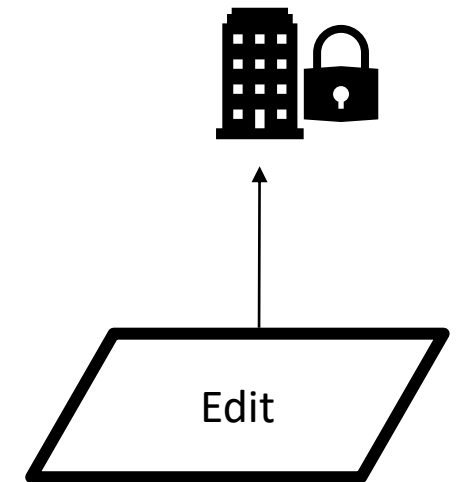
- **Purpose published**
 - **Read-only**
 - **Protected**
 - **Maintenance**
- Registered & Federated



Public – read only
Minimal attributes
No Federal property
No other jurisdictions



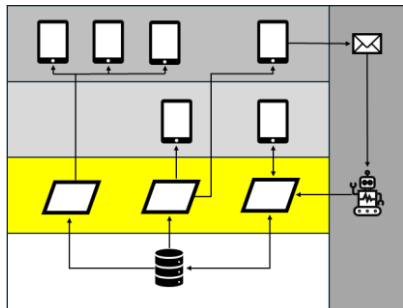
Protected – read only
All attributes
Includes non-standard
water sources



Internal – full edit control

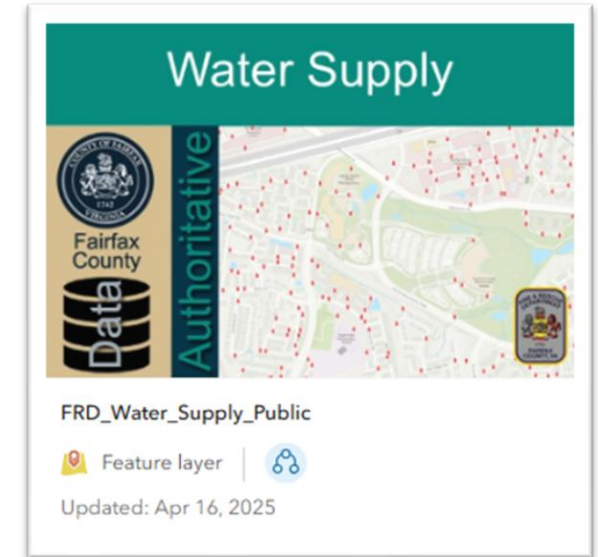
Water Supply Publishing

- Purpose published
 - Read-only
 - Protected
 - Maintenance
- **Registered & Federated**

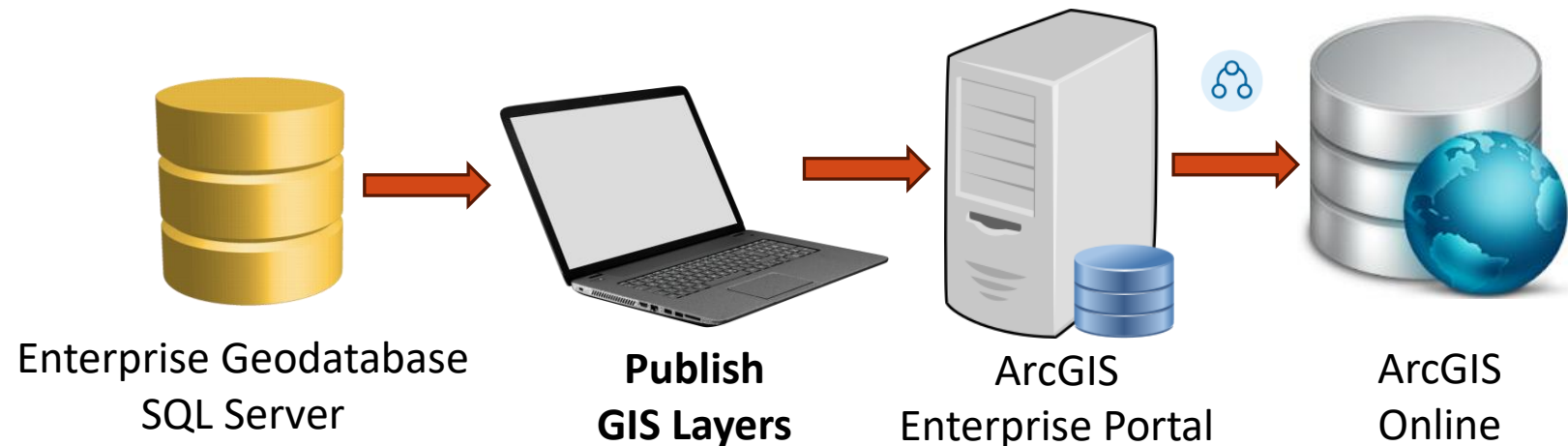


▼ Data and Layer Type ⓘ
 Reference registered data
☒ Map Image
☒ Feature
☐ Vector Tile
☒ Map Image and Feature
 Copy all data
☐ Feature
☐ Tile
☐ Map Image
☐ Vector Tile

Publishing options



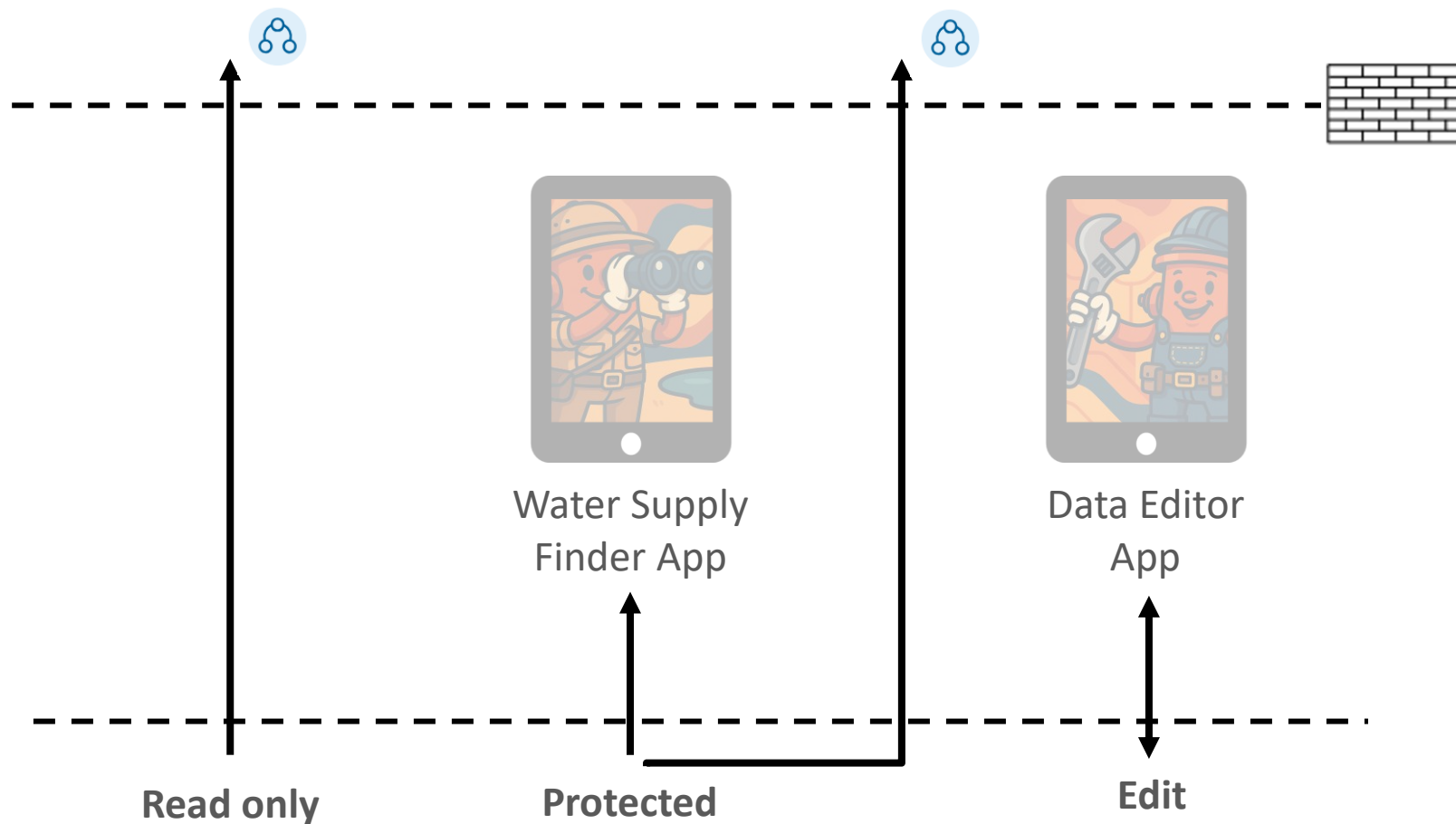
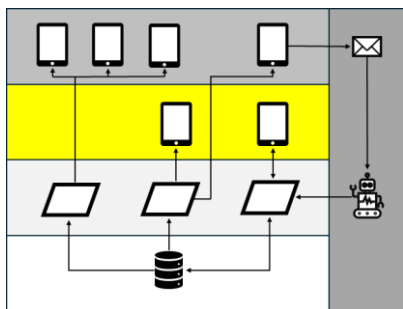
Authoritative Item





Water Supply Portal

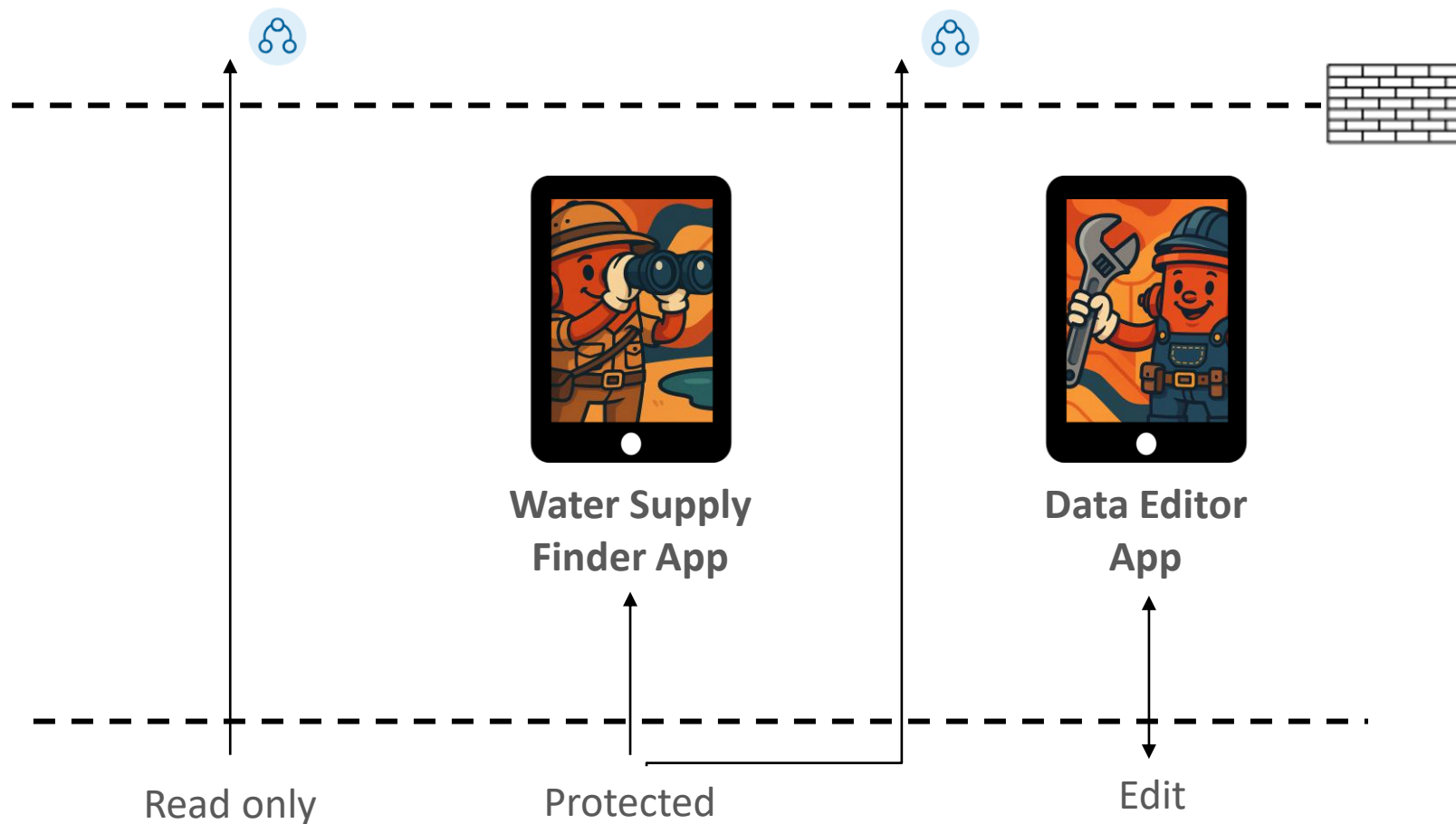
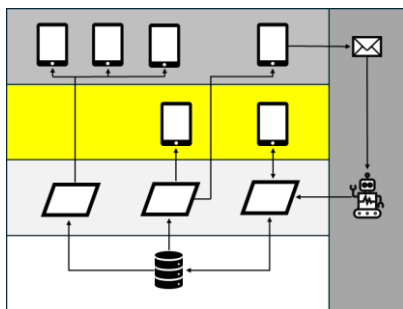
- **Federated Services**
 - Protected
 - Editable
 - Collaborations
- Applications
 - Water Supply Finder
 - FRD Data Editor





Water Supply Portal

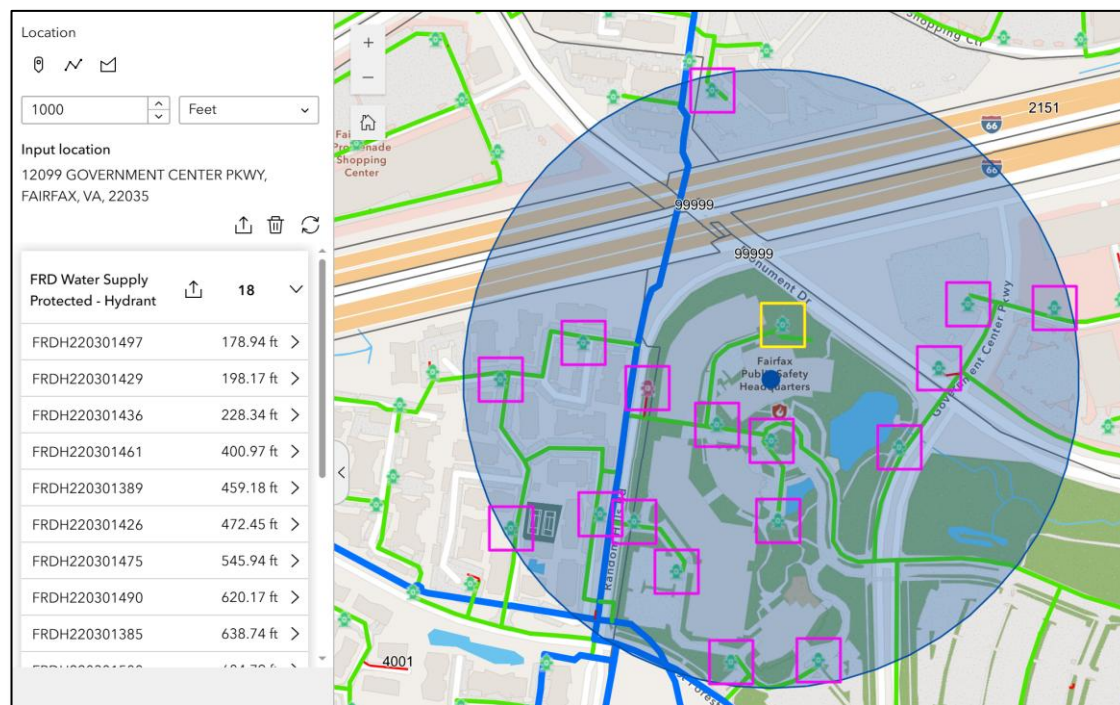
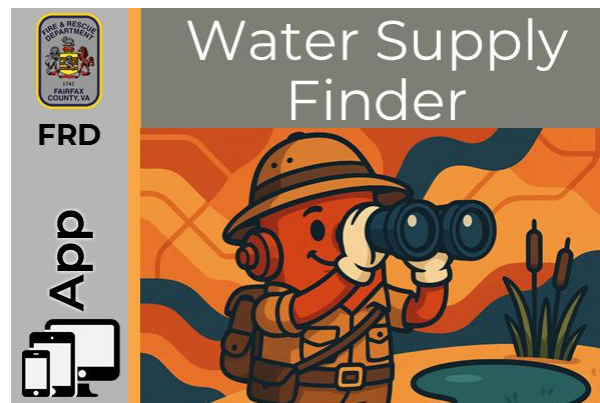
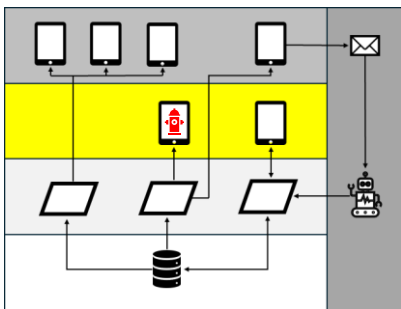
- Federated Services
 - Protected
 - Editable
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 - Water Supply Finder
 - FRD Data Editor





Water Supply Portal

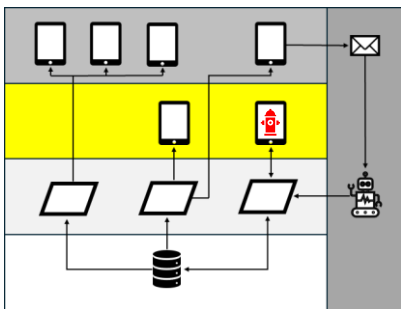
- Application
 - **Water Supply Finder**
 - Internal staff
 - Preplanning
 - Inspections
 - Inventory view
 - Water pipelines





Water Supply Portal

- Applications
 - **FRD Data Editor**
 - Real time editing
 - Attribute Rules apply
 - FRD GIS staff only



Fairfax County Fire & Rescue Department
Water Supply Editor

Experience Builder Web AppBuilder FDSU Google Maps

Feature Count: 32,380
FDSU_HydrantIssues_20250711

FRD Water Supply Out of Service

FRD Water Supply

- Access Door
- Access Hole
- Draft Site
- Dry Hydrant
- Hydrant Unknown Flow Rate
- Hydrant Modeled Flow Rate 100
- Hydrant Modeled Flow Rate 500
- Hydrant Modeled Flow Rate 1000
- Hydrant Modeled Flow Rate 1500
- Interstate Standpipe
- Tank
- others

FRDH221240543

Zoom to

Location Details

ZIP Code: 22124

Jurisdiction: FAIRFAX COUNTY

CAD Code: **HYDRANT: ACTIVE**

In First Due Size Up: Yes

Data Source

Source ID: H-00371-017

Source Type: HYDRANT

Source Status: Active

Ownership: Fairfax Water

Maintained By: FAIRFAX WATER

Data Steward: FAIRFAX CO FIRE

FRD Water Supply

Water Supply Type is: - All -

Water Supply Subtype is: - All -

Status is: - All -

Modeled Flow Rate is: - All -

Jurisdiction is: - All -

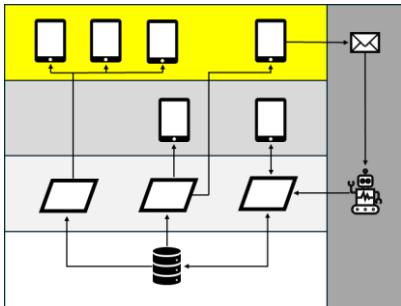
In First Due Size Up is: - All -

Maintained By is: - All -

FRD Hydrant ID	Water Supply Type	Water Supply Subt...	Status	Modeled Flow Rate	First Due	Fire Box	Address	Location Description	Location
FRDH221910442	Hydrant	Standard	Active	0	512	051219			
FRDH221910452	Hydrant	Air Release	Active	0	512	051202			

Water Supply ArcGIS Online

- Public Applications
 - Street Drills
 - Station Locator
 - Adopt a Hydrant
- Protected Applications
 - First Due Size Up

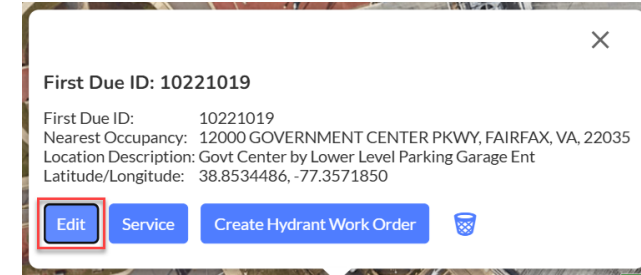


Public

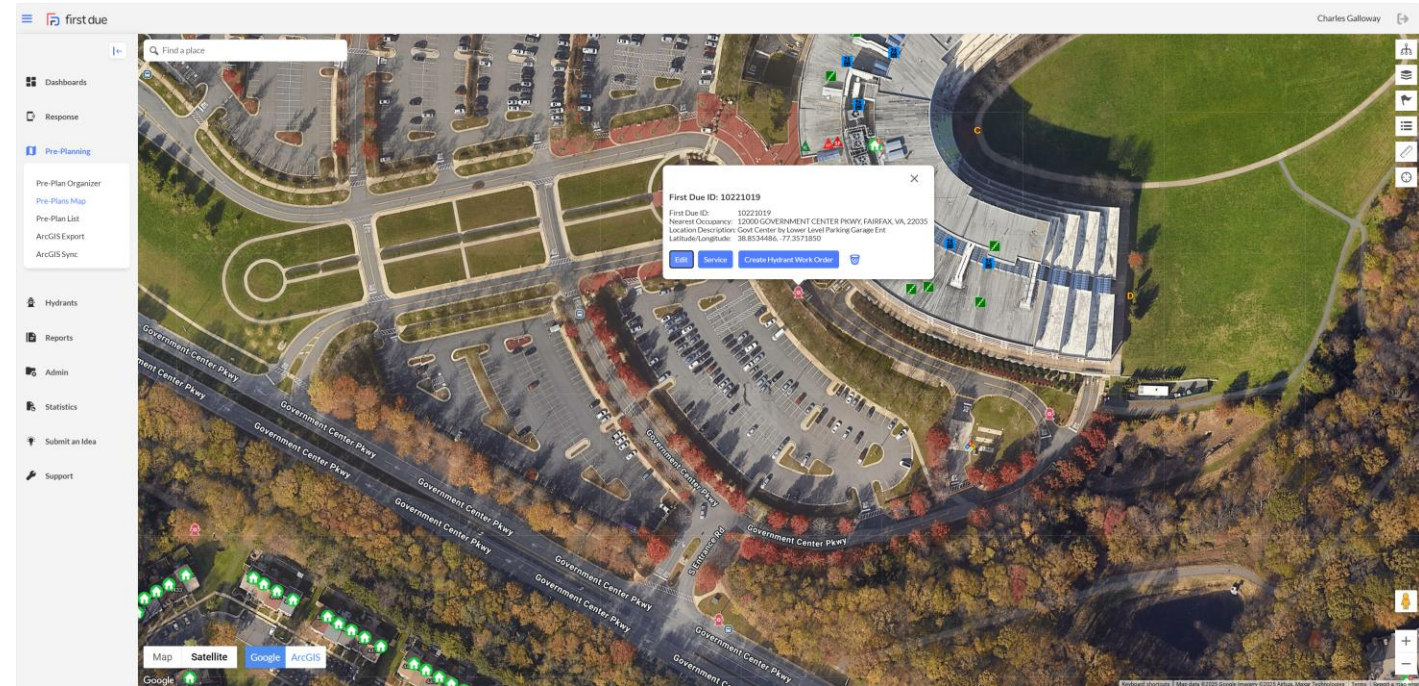
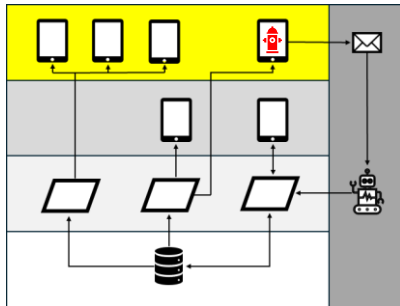


Protected

Water Supply ArcGIS Online



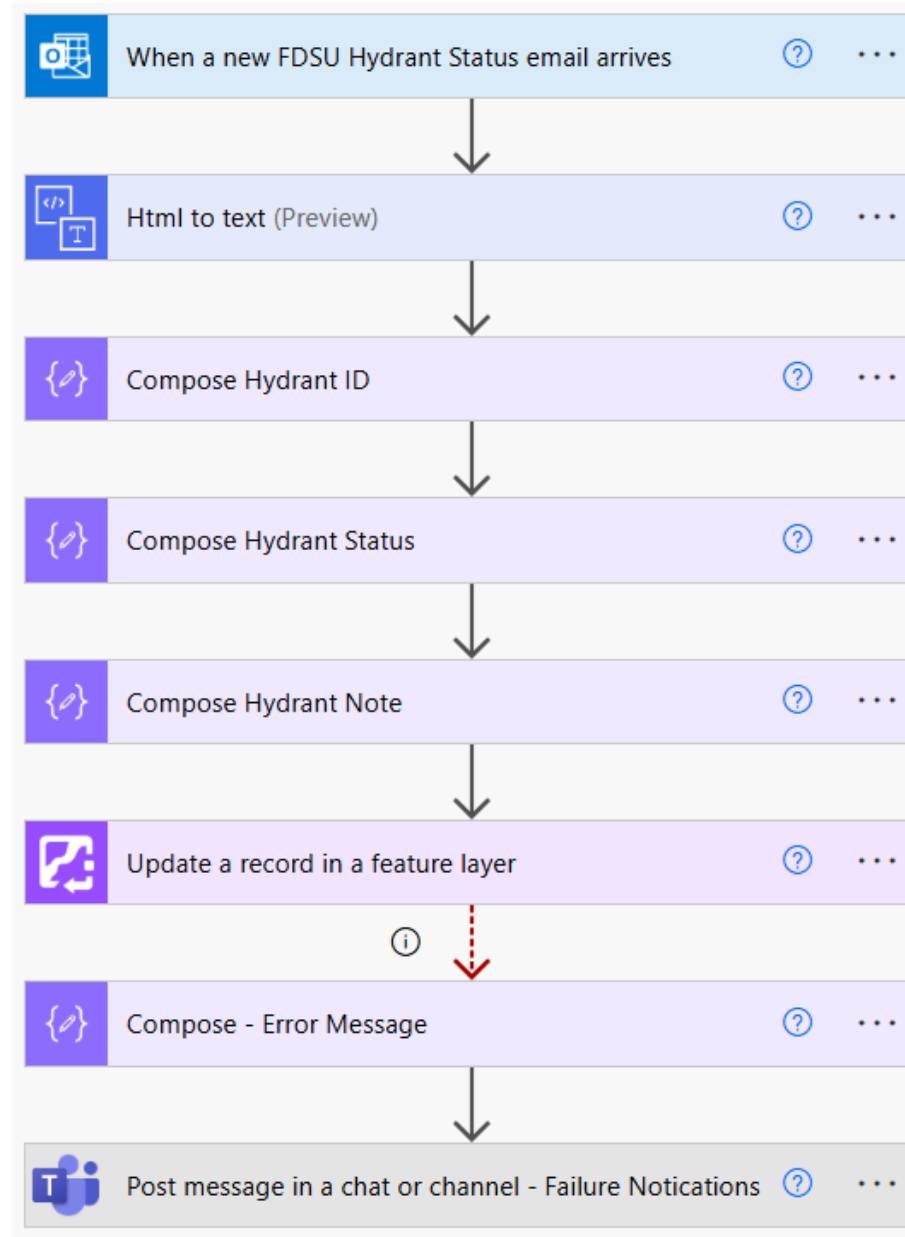
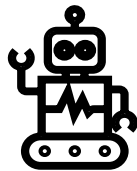
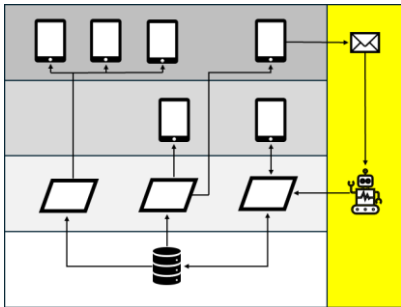
- Applications
 - **First Due Size Up**
 - 3rd party app
 - Displays units & incidents
 - Hydrant data syncs daily
 - Hydrant operational status



First Due Size Up desktop view

Water Supply Automation

- Power Automate
 - **FDSU Status Change**
 - CAD Notifications
 - Adopt a Hydrant

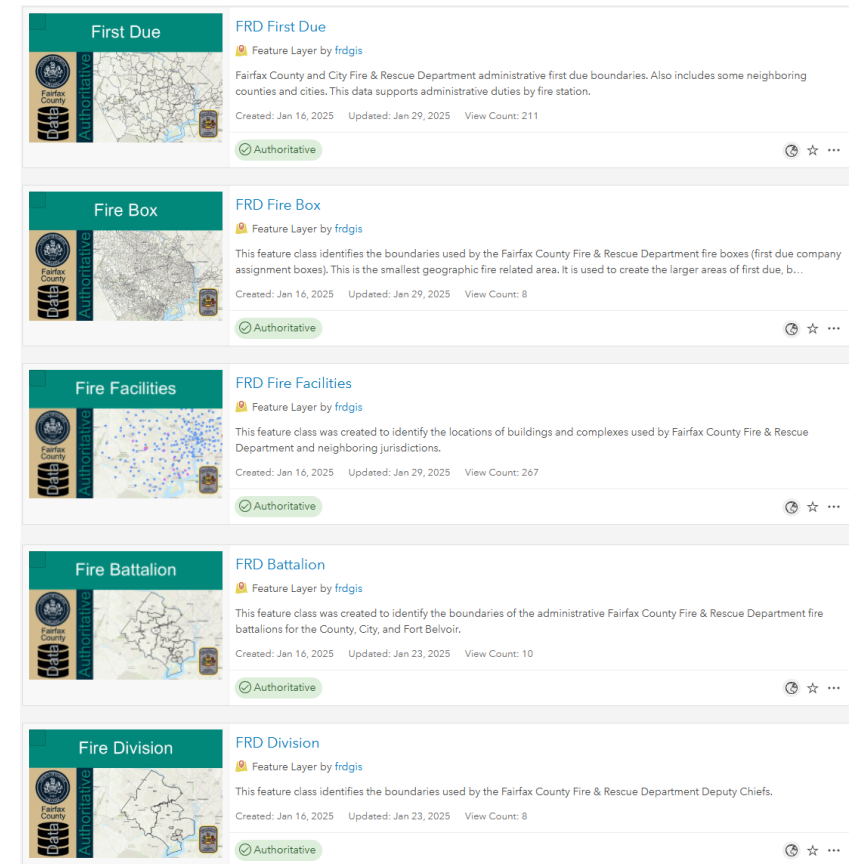


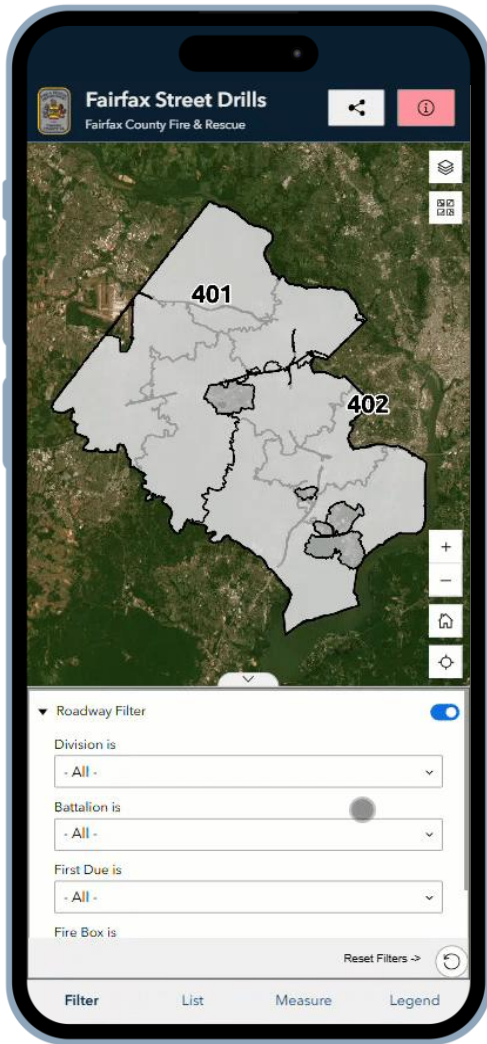
Authoritative Data Federated Services & Collaborations

■ Data view of authoritative GIS layers

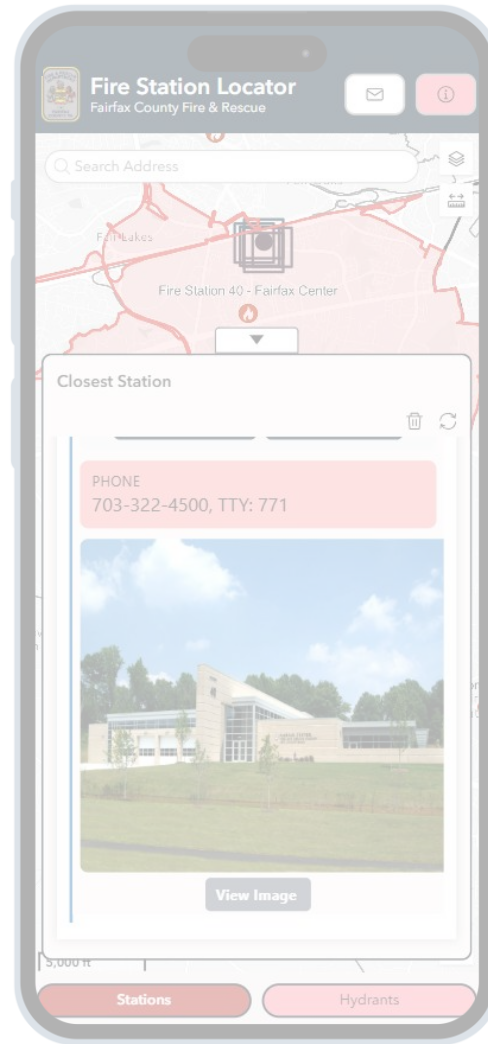
Benefits

- Centralized Access to Authoritative Data
- Secure, Scalable Sharing
- Available in Web & Mobile Applications, and more
- Real-Time Updates & Synchronization

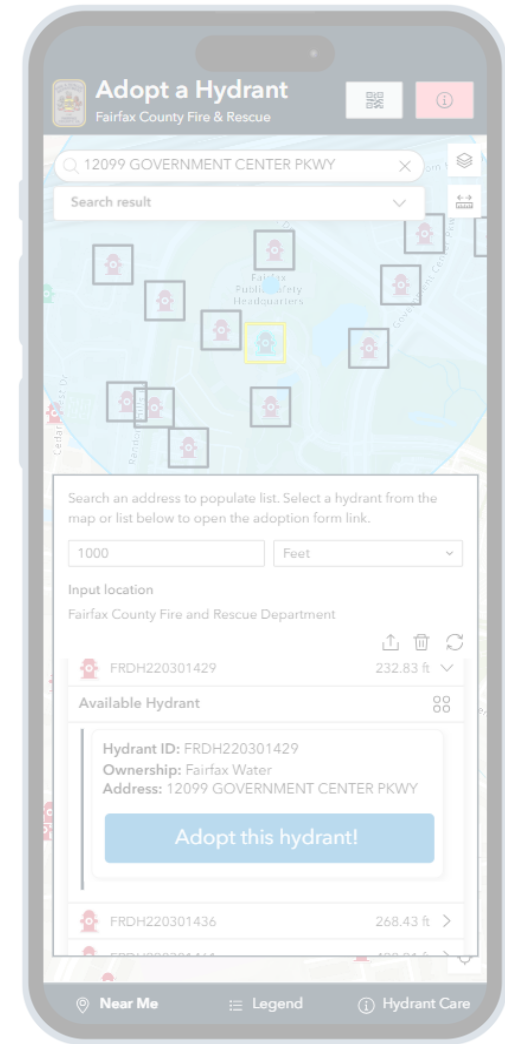




Street Drills



Fire Station Locator

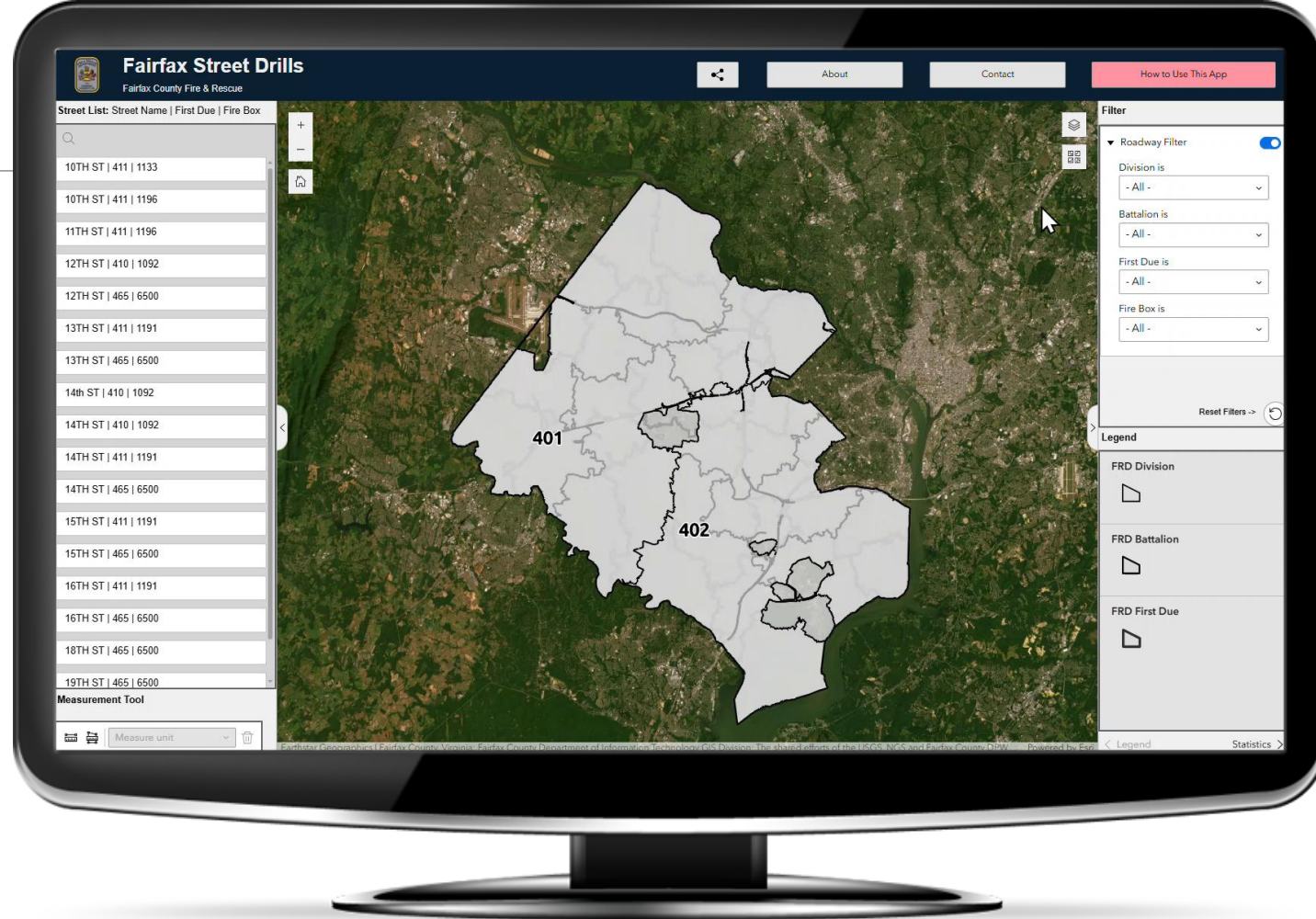


Adopt a Hydrant



Street Drills Purpose

- Training tool for first responders to learn streets and fire boxes
- Built using authoritative Fire & Rescue data
- Public-facing application with no login required
- Accessible on any device for quick, on-demand use

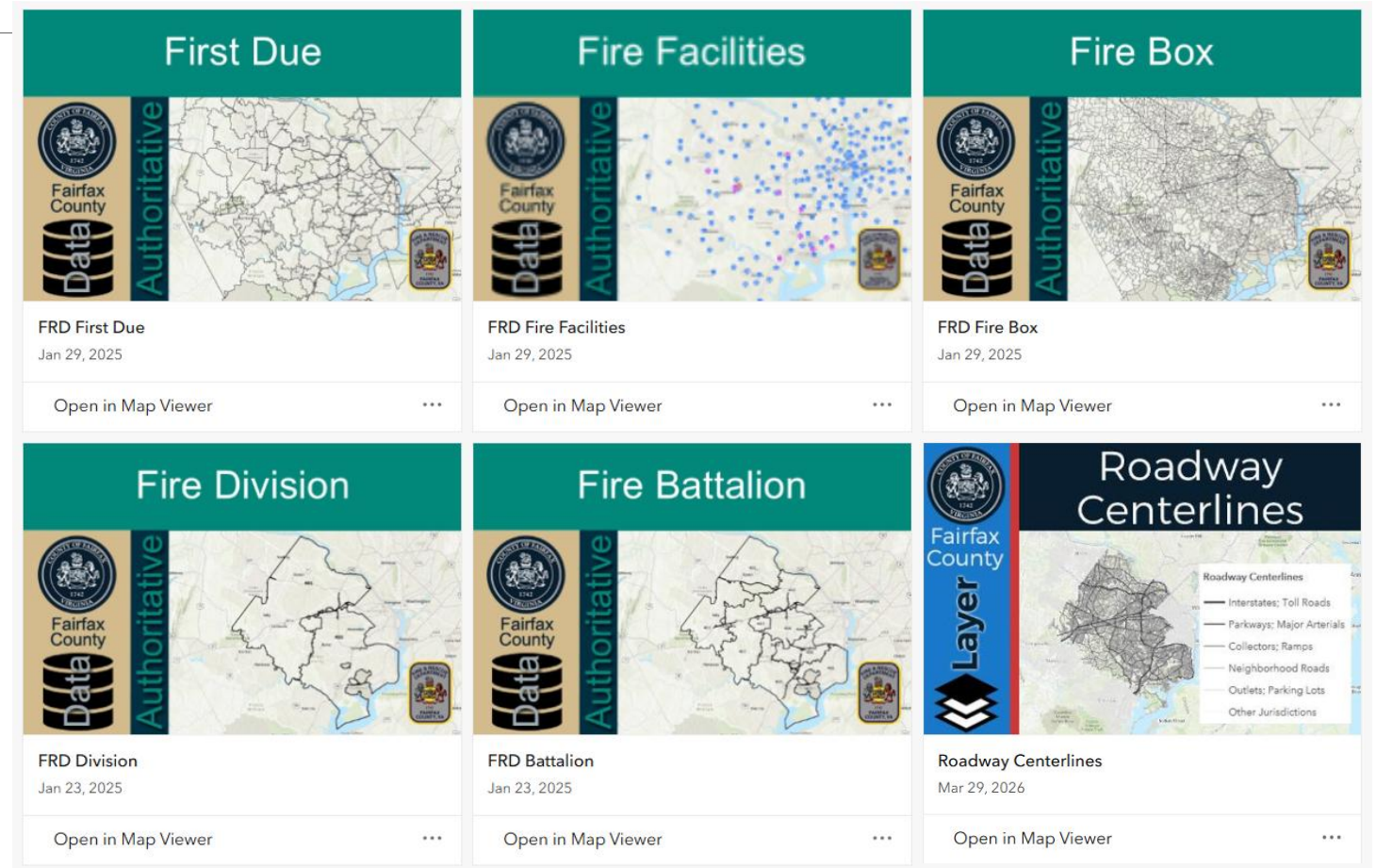


Street Drills Data Sources

- Authoritative internal Fire & Rescue datasets (first due, fire box boundaries)
- External dataset: Fairfax County streets layer
- Streets layer is clipped by fire box boundaries to drive filtering
- Integrated hydrant data for added operational context

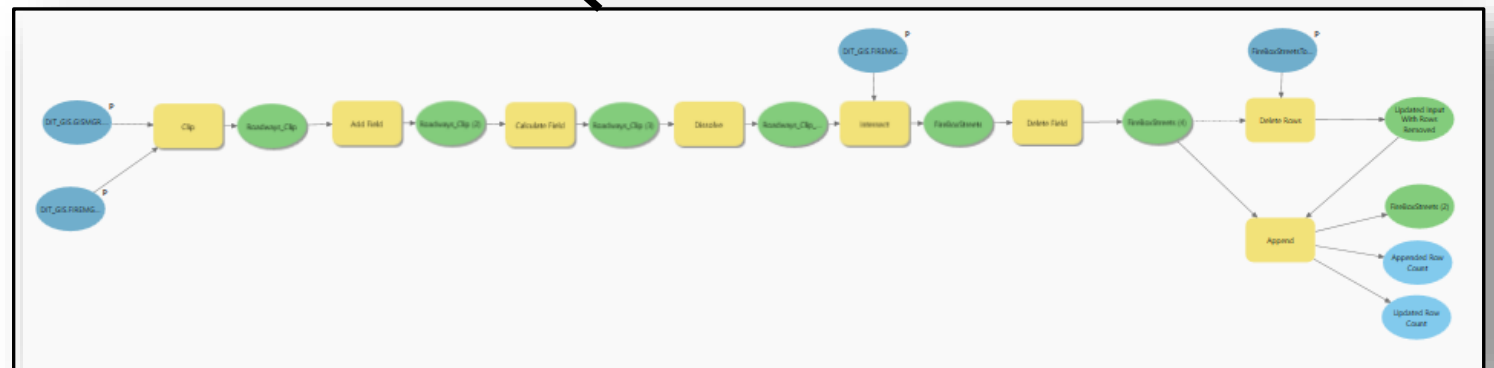
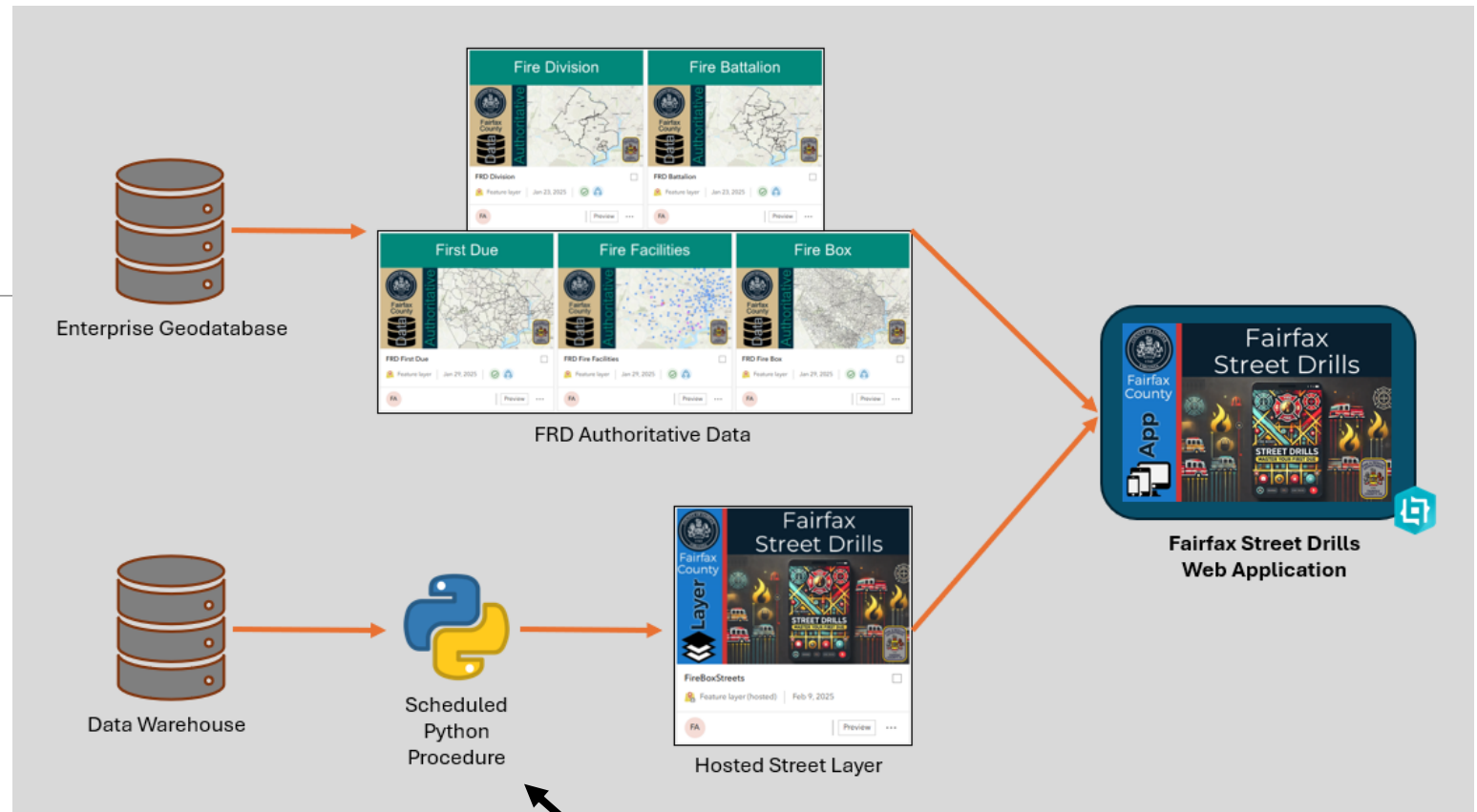


Street Drills



Street Drills Automation

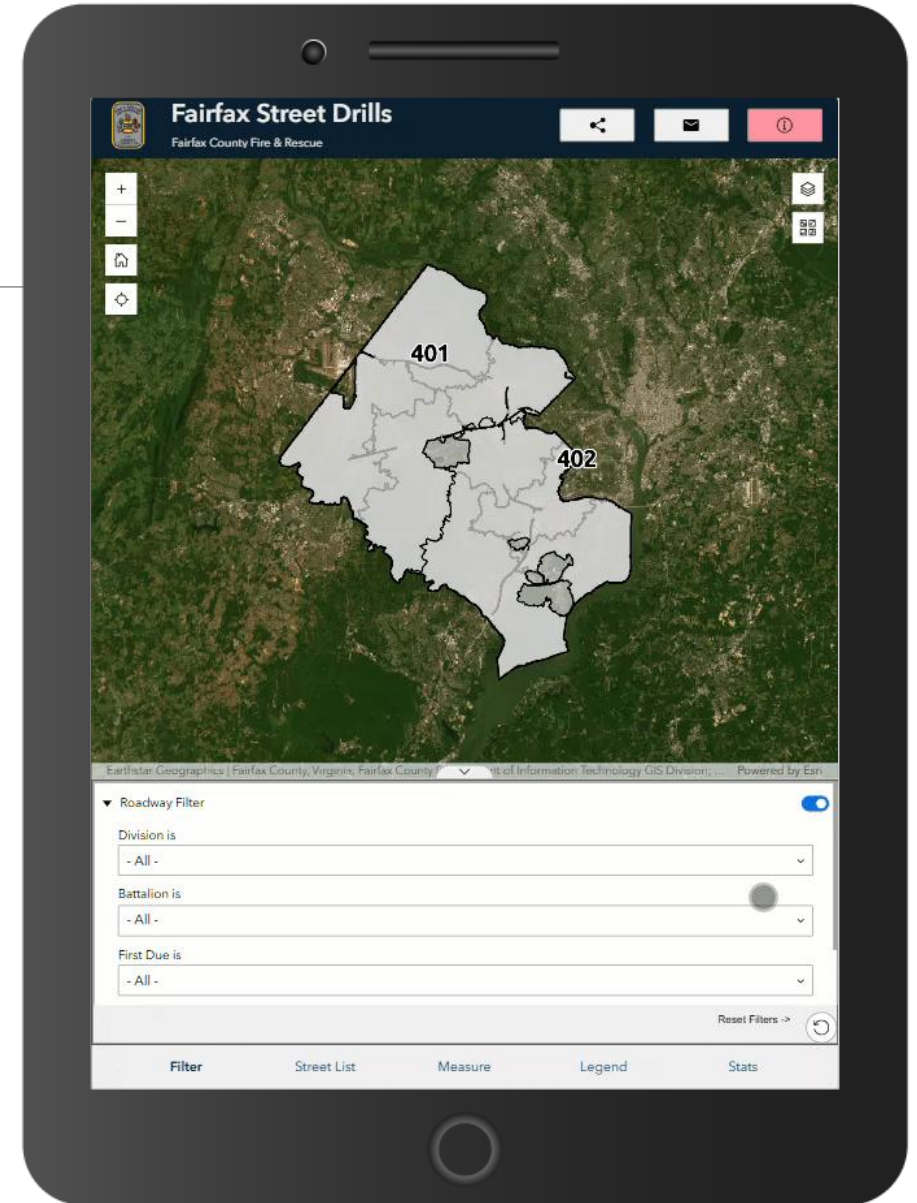
- Automated workflows to clip streets by fire box boundaries
- Scheduled processing ensures data stays up to date
- Reduces manual data prep and maintenance effort
- Supports consistent and repeatable data outputs for the app



Street Drills

Core Functionality

- Filter by first due or fire box
- View list of labeled and unlabeled streets for study
- Click a street to highlight it on the map
- Measure tool for preplanning and distance awareness
- On-demand training
- Export street lists to CSV for offline use

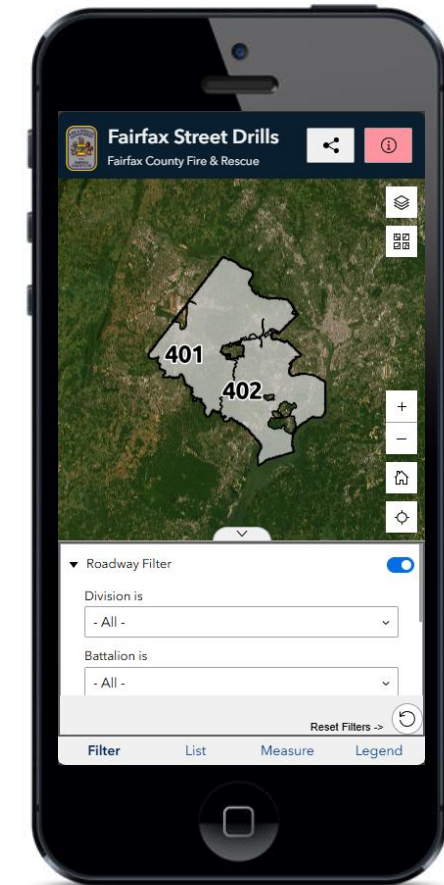


Street Drills

User Interface & Accessibility



- Simple, intuitive interface designed for quick interaction
- Prominent, reactive buttons for filtering and navigation
- Optimized for desktop, tablet, and mobile devices
- QR code access for quick launch in the field
- Contact info readily available for feedback and support
- Statistics on roadway counts by division, battalion, first due, and fire box



Street Drills

How to Use



■ Step 1: Filter Roadways

The screenshot shows the Fairfax Street Drills web application. The header includes the Fairfax County Fire & Rescue logo, the title 'Fairfax Street Drills', and navigation links for 'About', 'Contact', and 'How to Use This App'. The main interface is divided into three sections: a 'Street List' on the left, a central map, and a 'Filter' panel on the right. The 'Street List' has a search bar and a table of streets with columns for 'Street Name', 'First Due', and 'Fire Box'. The map shows a satellite view of a region with two areas highlighted in grey and labeled '401' and '402'. The 'Filter' panel includes a 'Roadway Filter' section with a toggle switch and three dropdown menus for 'Battalion is', 'First Due is', and 'Fire Box is'. Below the filter is a 'Legend' section with three items: 'FRD Division', 'FRD Battalion', and 'FRD First Due'. At the bottom of the filter panel is a 'Statistics' section showing counts for 'Battalion count: 10', 'First Due count: 45', 'Fire Box count: 1,123', and 'Roadway count: 20,878'. A 'Measurement Tool' is located at the bottom left of the map area.



Street Drills

How to Use



- Step 2: Use Roadway List to Highlight Streets on the Map

Fairfax Street Drills
Fairfax County Fire & Rescue

Street List: Street Name | First Due | Fire Box

Search

- BLACK CT | 414 | 1408
- CLARIDGE CT | 414 | 1408
- COMMONWEALTH BLVD | 414 | 1408
- CONSTABLE CT | 414 | 1408
- DALROY CT | 414 | 1408
- DUNDALK ST | 414 | 1408
- EASTLAKE DR | 414 | 1408
- ELLINGTON CT | 414 | 1408
- ENFORD CT | 414 | 1408
- GAINSBOROUGH DR | 414 | 1408
- LATIMER CT | 414 | 1408
- LAVERY CT | 414 | 1408
- LLEWELLYN CT | 414 | 1408
- ORCHARDSON CT | 414 | 1408
- PUMPHREY DR | 414 | 1408

Measurement Tool

Measure unit: [v]

Filter

Roadway Filter [on]

Battalion is: - All -

First Due is: 414

Fire Box is: 1408

Legend

FRD Fire Facilities

- Federal
- Facility
- Headquarters
- Station

FRD Fire Box

Statistics

Battalion count: 1
First Due count: 1
Fire Box count: 1
Roadway count: 20



Street Drills

Street Drills How to Use



■ Step 3: Search Street List

The screenshot shows the 'Fairfax Street Drills' web application. The header includes the Fairfax County Fire & Rescue logo, the title 'Fairfax Street Drills', and navigation links for 'About', 'Contact', and 'How to Use This App'. The main interface is divided into three sections: a search list on the left, a map in the center, and a filter/legend/statistics panel on the right.

Search List: A table with columns for Street Name, First Due, and Fire Box. The list includes streets such as ABERCROMBY CT, ADAMS HILL RD, ADMIRALS HILL CT, ALDENHAM LN, ALEXANDER BELL DR, AMBERJACK CT, AMERICAN DREAM WAY, APPROACH LN, APRICOT CT, ARROW RD, ASCOT WAY, ASSOCIATION DR, BACHAN CT, BALD EAGLE WAY, and BANDIT LOOP.

Map: A map of Fairfax County showing street networks. Specific areas are labeled with numbers 425, 431, and 442.

Filter Panel: Contains a 'Roadway Filter' section with dropdown menus for 'Battalion is' (set to '- All -'), 'First Due is' (set to '425'), and 'Fire Box is' (set to '- All -').

Legend: Shows symbols for 'FRD First Due' (a black square) and 'FRD Fire Box' (a purple square). It also lists 'Roadway Centerlines'.

Statistics: A summary of counts: Battalion count: 1, First Due count: 1, Fire Box count: 23, and Roadway count: 444.

At the bottom of the map, there is a footer with attribution: 'Maxar | Fairfax County, Virginia | Fairfax County, Virginia; Fairfax County Fire & Rescue Department, Data Analytics Strategy Management Division' and 'Powered by Esri'.



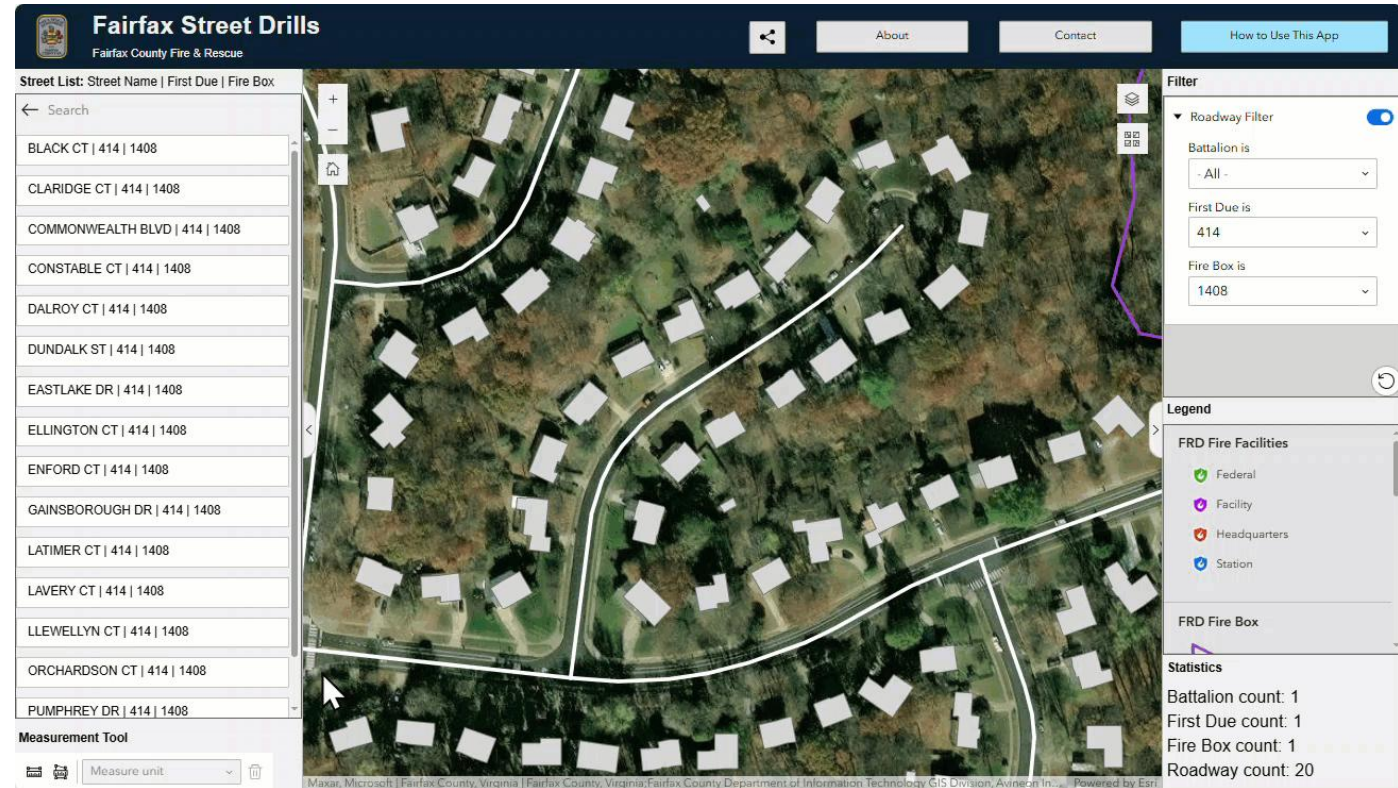
Street Drills

Street Drills

How to Use



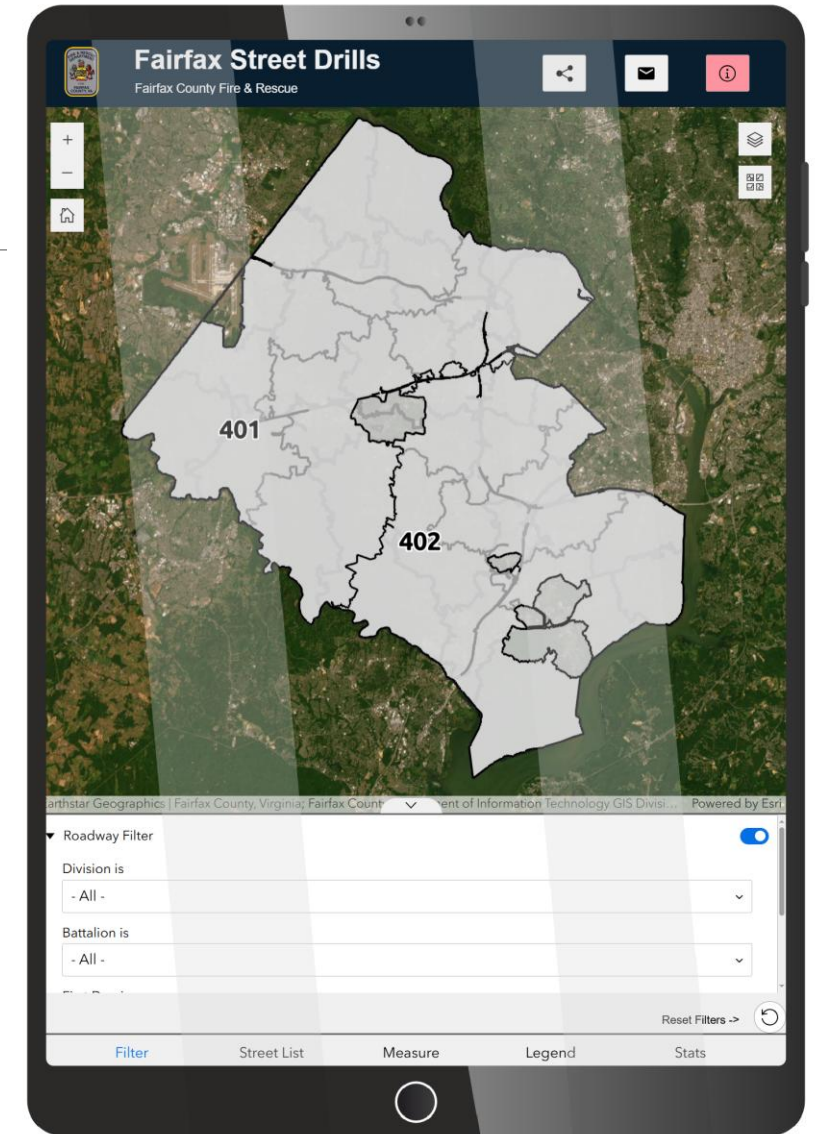
- Step 4: Measure as a preplan tool

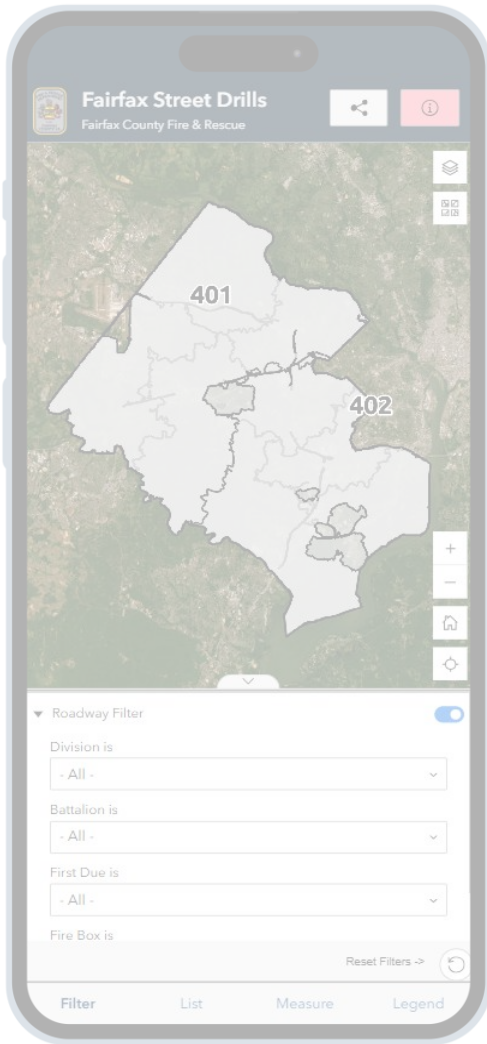


Street Drills

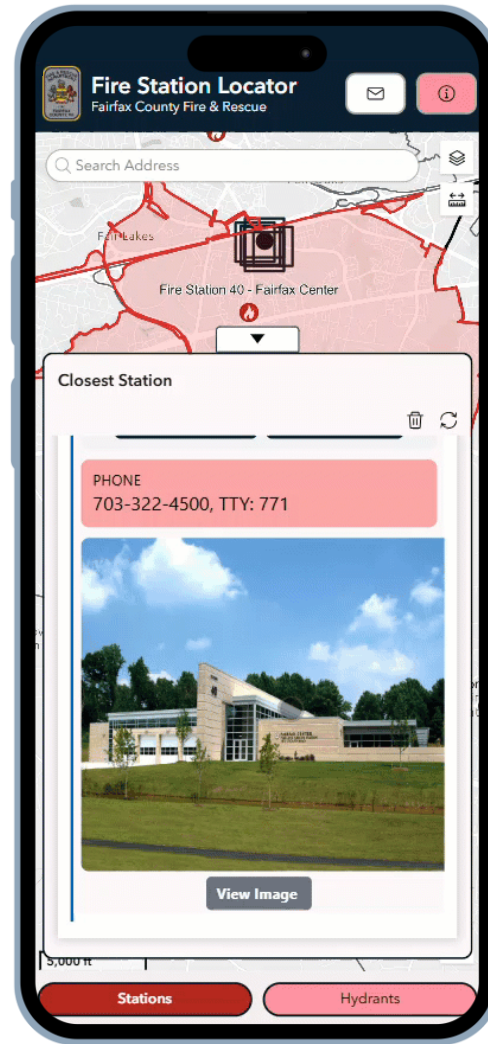
Why Public-Facing Matters

- Eliminates login barriers for first responders
- Accessible on personal devices without county credentials
- Increases usability and real-world usage
- Maintains security by limiting sensitive data exposure

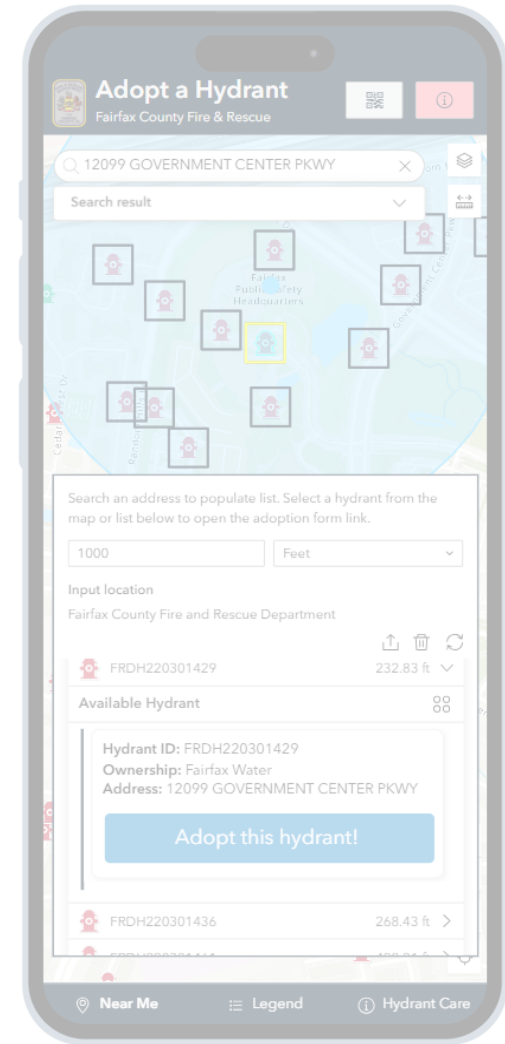




Street Drills



Fire Station Locator



Adopt a Hydrant

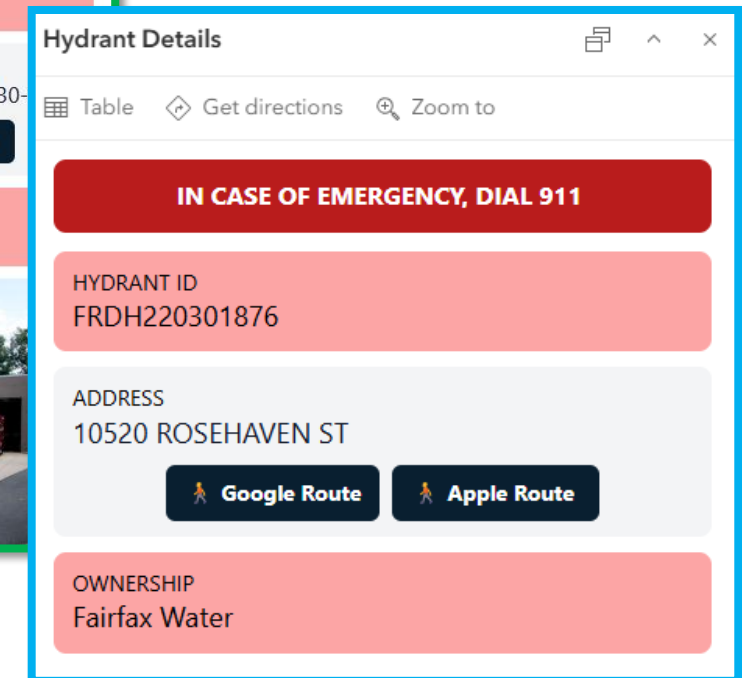
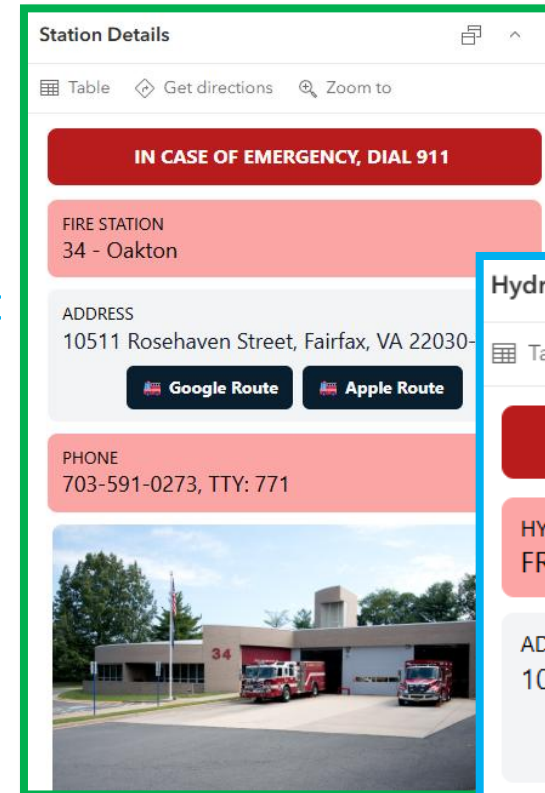


Station Locator

Fire Station Locator Overview



- Enables residents to quickly find their responding fire station and nearby fire hydrants
- Provides key details for each **station** (address, phone number, image) and **hydrant** (address, hydrant ID, ownership)
- Increases public awareness of fire/EMS coverage, response areas, and available services
- Accessible from the Fire & Rescue Department section of the Fairfax County website



Station Locator

Fire Station Locator Overview

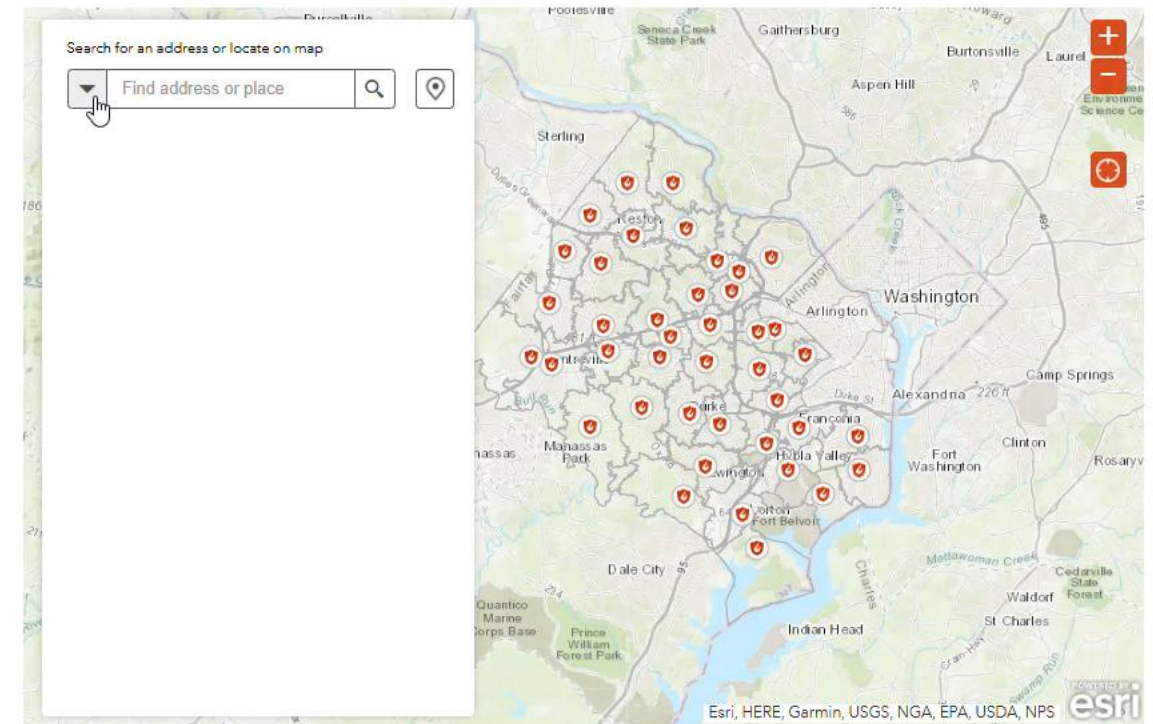


- Replaces the legacy **Fire Station Locator** built in Web App Builder
- Rebuilt in **ArcGIS Experience Builder** due to end of Web App Builder support for updates
- Ensures the app remains secure, maintainable, and compatible with future platform enhancements
- Provides a more modern, responsive user experience on both desktop and mobile devices



Station Locator

Fire Station Locator



Fire Station Locator

The Data



- Uses a **joined layer view** of fire stations table and their corresponding first-due response areas

- Additional Layers:

- Fire Station Points
- Fire Hydrant Points
- Address Points
- Fairfax County Boundary

First Due

45 records, 0 selected

length	FIRST_DUE_INT	FIRST_DUE_TEXT	STATION_NUMBER	STATION_NAME	PICTURE	PHONE	ADDRESS
	401	401	1	Mclean	View	703-356-6671, TTY: 771	1455 Laughlin Avenue, M...
	402	402	2	Vienna	View	703-938-2242, TTY: 771	400 Center Street SW, Vie...
	404	404	4	Herndon	View	703-437-1233, TTY: 771	680 Spring Street, Herndo...
	405	405	5	Franconia	View	703-971-5858, TTY: 771	6300 Beulah Street, Alexa...
	408	408	8	Annandale	View	703-256-2552, TTY: 771	7128 Columbia Pike, Anna...
	409	409	9	Mount Vernon	View	703-780-0150, TTY: 771	2601 Sherwood Hall Lane,...
	410	410	10	Bailey's Crossroads	View	703-820-2345, TTY: 771	3601 Firehouse Lane, Falls...
	411	411	11	Penn Daw	View	703-765-4404, TTY: 771	6624 Hulvey Terrace, Alex...
	412	412	12	Great Falls	View	703-759-2300, TTY: 771	9916 Georgetown Pike, G...
	413	413	13	Dunn Loring	View	703-560-1539, TTY: 771	2148 Gallows Road, Dunn ...
	416	416	16	Clifton	View	703-830-1901, TTY: 771	12645 Chapel Road, Clifto...
	418	418	18	Jefferson	View	703-573-4505, TTY: 771	3101 Hodge Place, Falls C...
	420	420	20	Gunston	View	703-339-5970, TTY: 771	10417 Gunston Road, Lort...
	421	421	21	Fair Oaks	View	703-591-0247, TTY: 771	12300 Route 50, Fairfax, V...
	423	423	23	West Annandale	View	703-978-3340, TTY: 771	8914 Little River Turnpike, ...

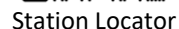


First due attribute table
now including station
details



Station Locator

- [illegible]

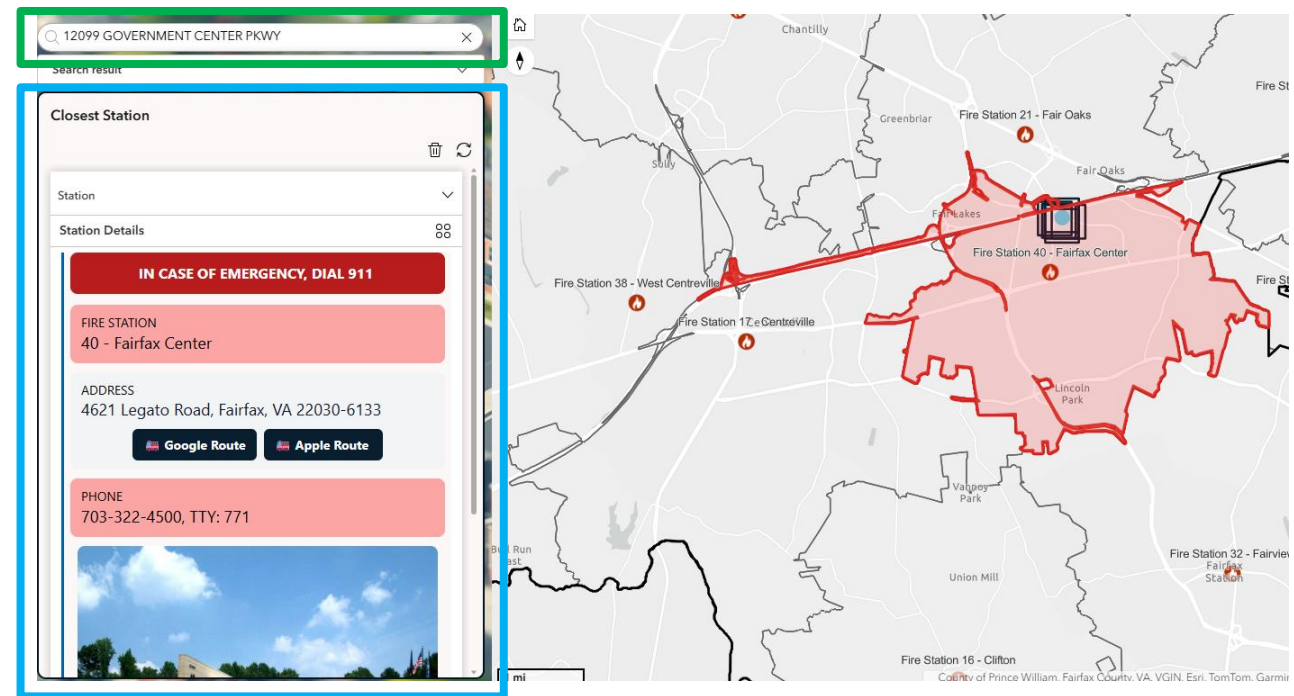


Fire Station Locator Experience Builder



- Built as an ArcGIS Experience Builder web app
- Leverages the **Near Me** widget for location-based results
 - **Stations:** Near Me identifies the first-due area containing the entered address, then returns the serving station and coverage area
 - **Hydrants:** Near Me returns the closest hydrants by proximity to the entered address (500 ft)
 - Users start by entering an address in the search widget, which triggers the Near Me widget to run its analysis

Search widget

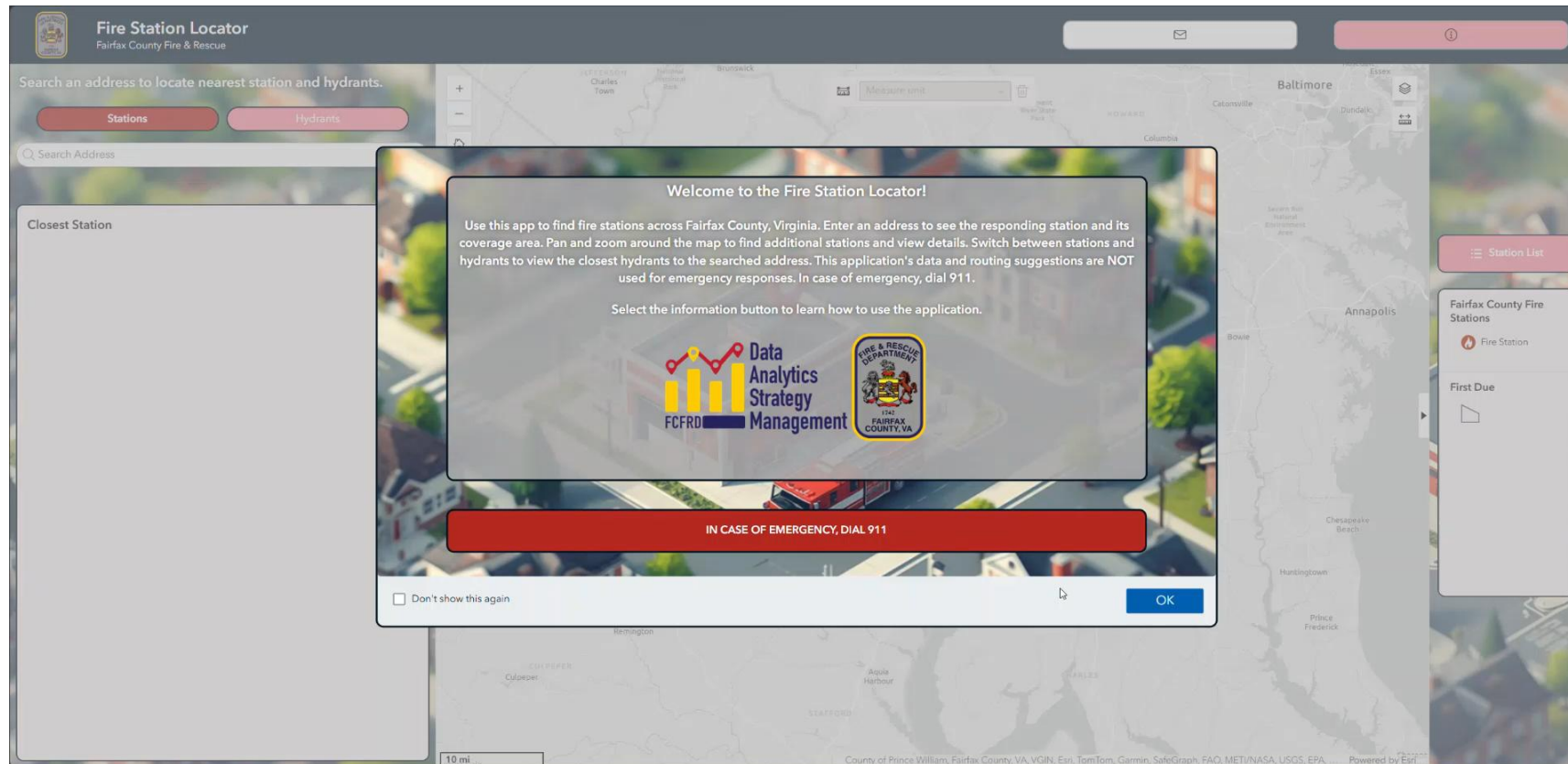


Near Me widget



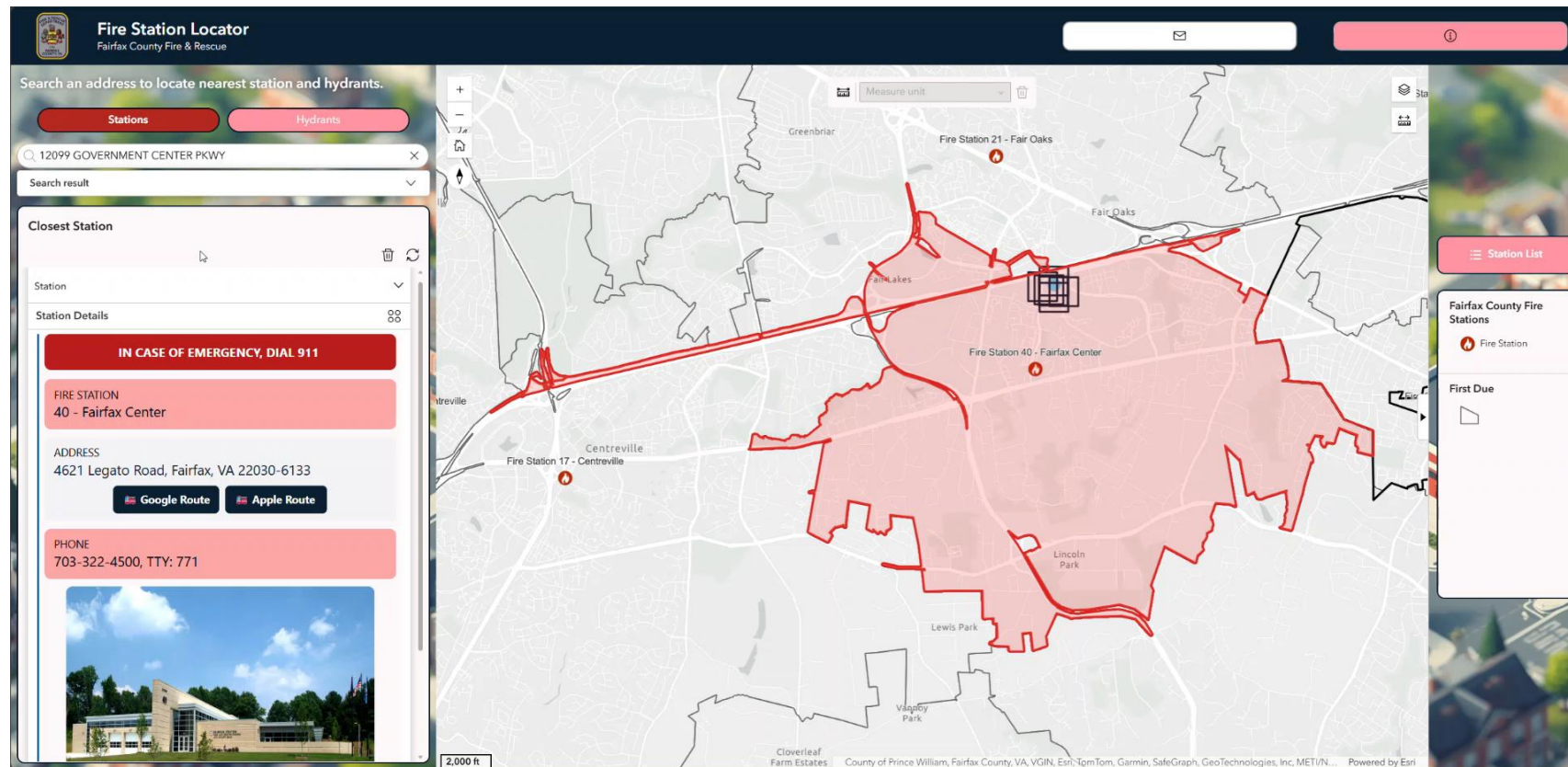
Station Locator

Fire Station Locator Station Details

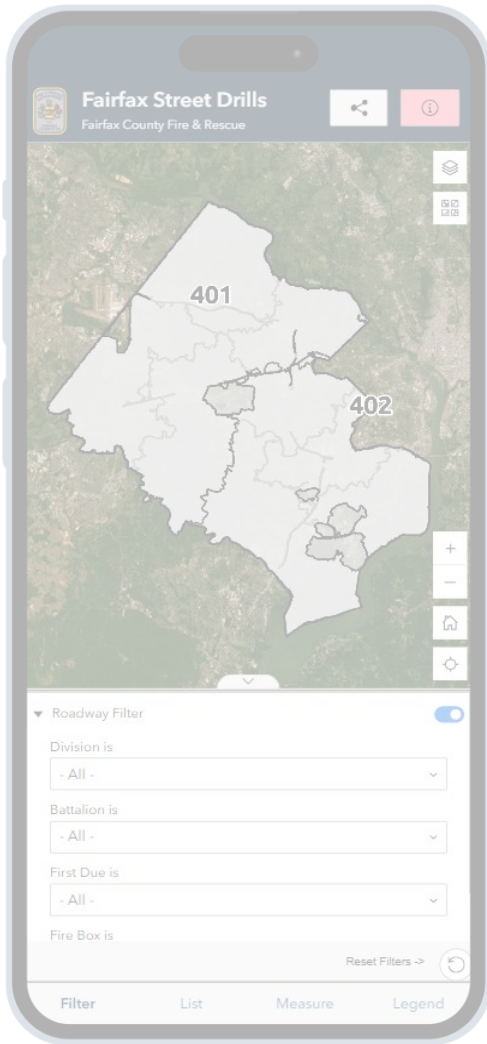


Station Locator

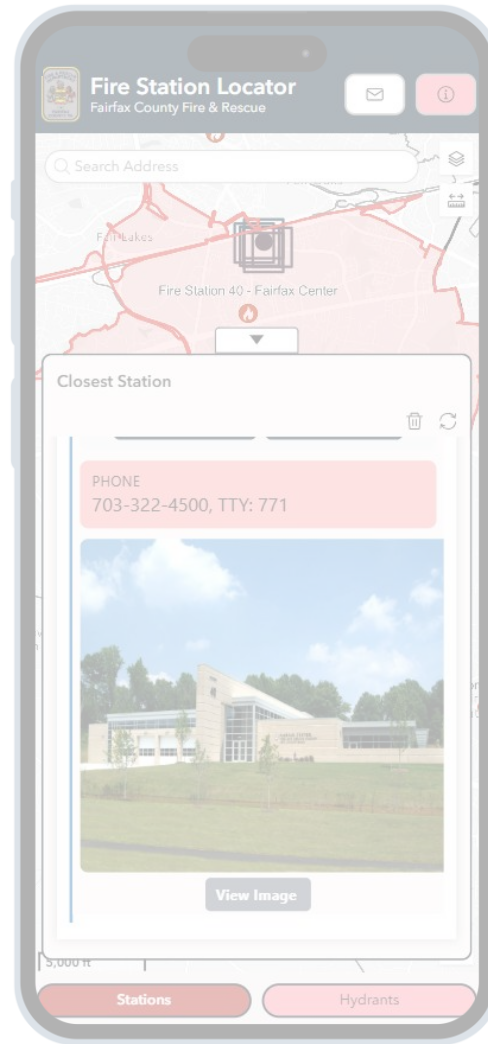
Fire Station Locator Hydrant Details



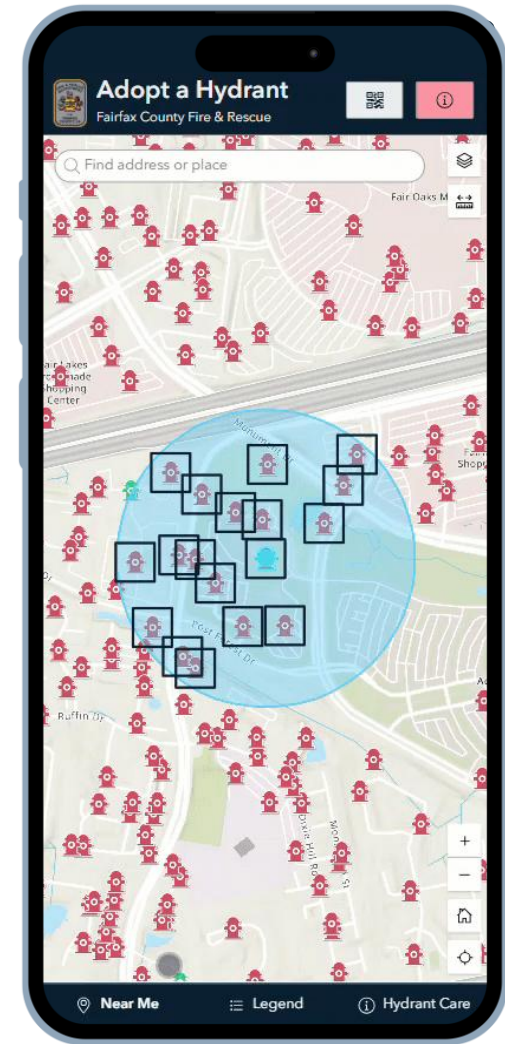
Station Locator



Street Drills



Fire Station Locator



Adopt a Hydrant



Adopt a Hydrant

- Overview
- Data
- Application Features
- How to Use
- Certificates
- Enhancements



Adopt a Hydrant Overview



■ Overview

- Saves seconds and improves operational efficiency
- Adopters pledge to keep hydrants clear year-round of snow or debris, providing vital support to the Fire & Rescue Department.
- Ensures responders can quickly locate and access hydrants during emergencies—especially in winter or overgrown conditions.

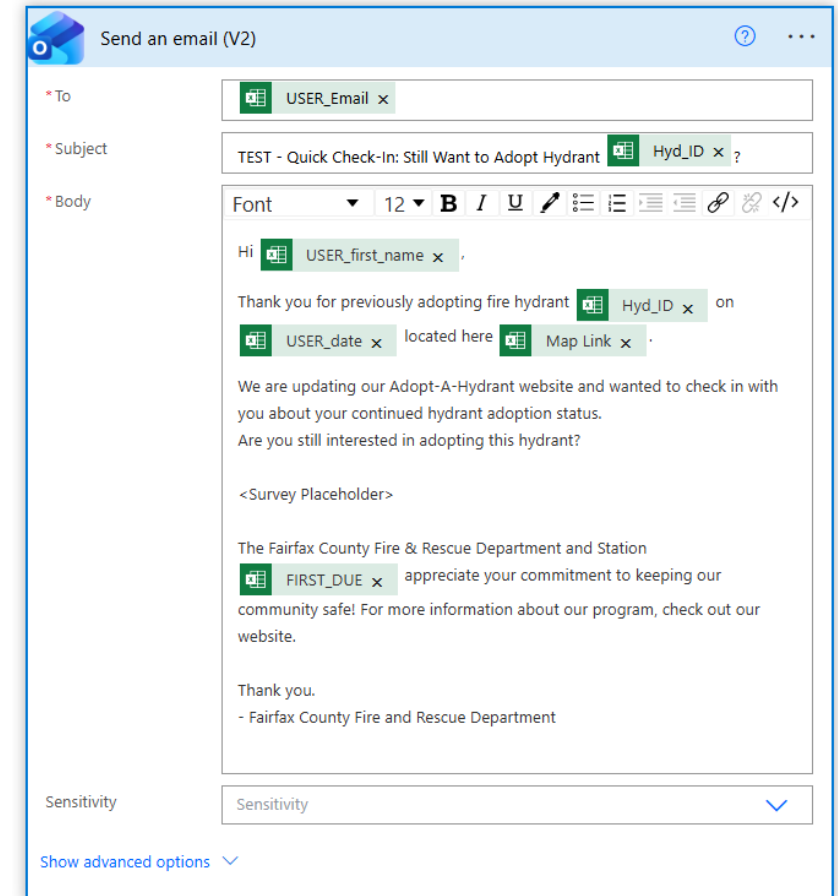


Adopt a Hydrant

Adopt a Hydrant Legacy Adoptions



- Program started 2017
- People take great pride in participating
- Power Automate
 - Import legacy table into excel
 - Variables mapped to Email & Survey123 URL parameters
 - Emailed all ~500 legacy adopters
 - 2 clicks to have early access and preserve your adoption status
- We wanted to preserve those who staked their claim
 - 2 weeks before announcing the updated program

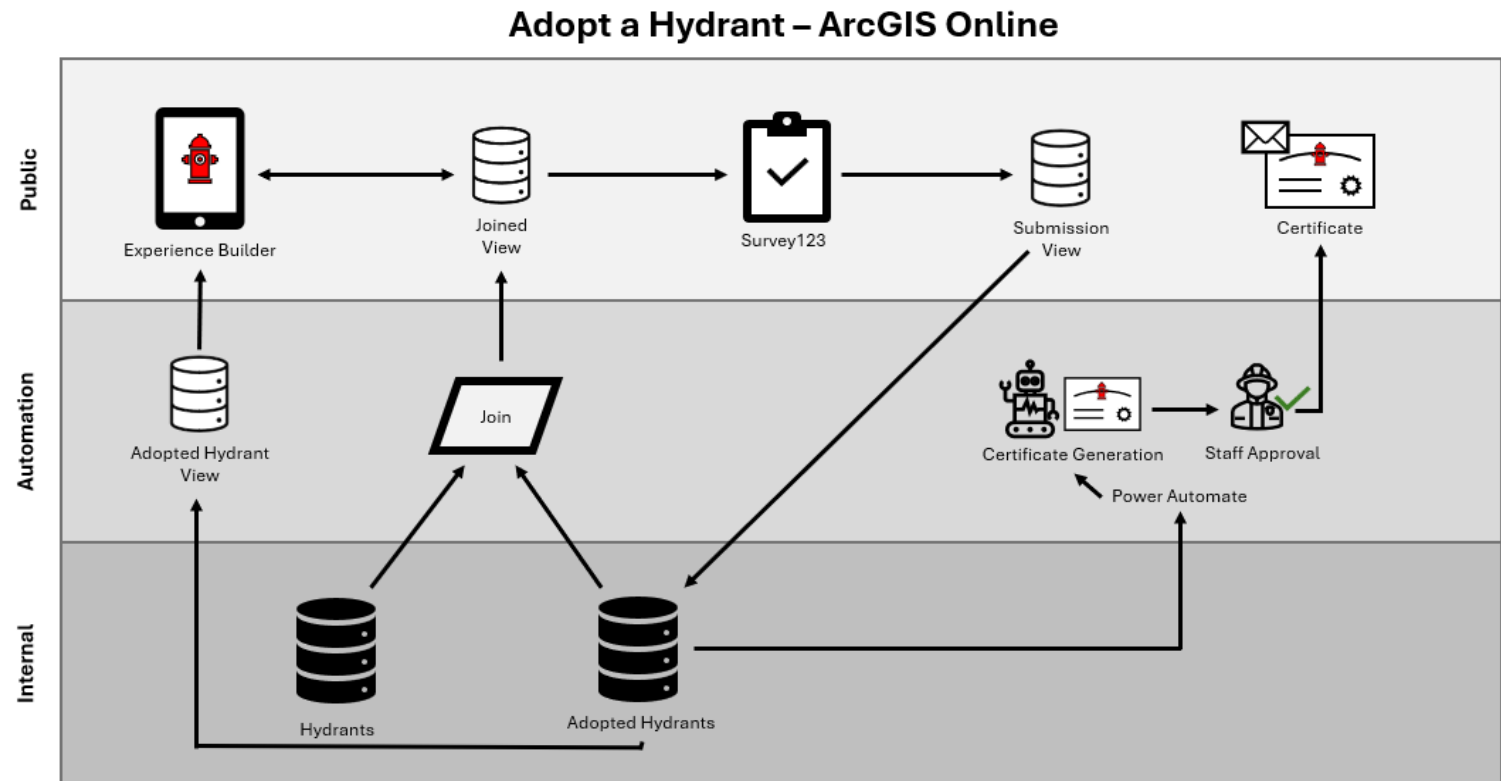


Adopt a Hydrant

Adopt a Hydrant Data



- Centralized source data
- Joined layer view of authoritative GIS layers
 - Water Supply + Adopted Hydrants
 - Filters based on if an attribute field is blank
 - Goal: to only show available hydrants to the user
- Using “views” of the data ensures security and consistency



Adopt a Hydrant

Adopt a Hydrant Survey123



- Survey123 Connect
 - Used to build adoption survey
 - More configurable than web environment

The screenshot shows the Survey123 Connect application interface, which is a spreadsheet-like environment for configuring a survey form. The interface includes a menu bar (File, Home, Insert, Page Layout, Formulas, Data, Review, View, Automate, Help, Foxit PDF, Table Design), a ribbon with various tool groups (Clipboard, Font, Paragraph, Alignment, Number, Styles, Cells, Editing, Sensitivity, Add-ins, Analyze Data), and a main grid with columns labeled A through H. The grid contains various data types and labels for survey questions, such as 'type', 'name', 'label', 'hint', 'guidance_hint', 'appearance', 'required', and 'required_message'. The bottom of the screen shows a status bar with 'Ready' and 'Accessibility: Investigate'.

The screenshot shows the 'Hydrant Adoption Form' interface. At the top, it says 'ADOPT HYDRANT' with the 'fairfax county fire and rescue department' logo. Below this, a message reads: 'Thank you for participating in the Adopt a Hydrant program! Please complete the form below to receive your official adoption certificate by email.' The form is divided into sections: 'Hydrant Location' (with a map showing the location at Crosspointe), 'Hydrant Address' (with the address 8514 HERON POND LN), and 'Hydrant ID' (with the ID FRDH220390047). A 'This is a read-only question.' message is displayed on the map.



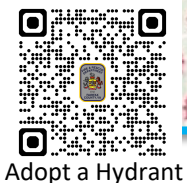
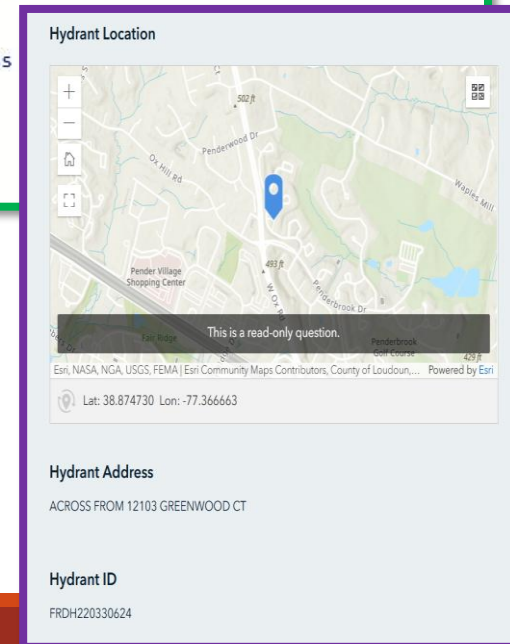
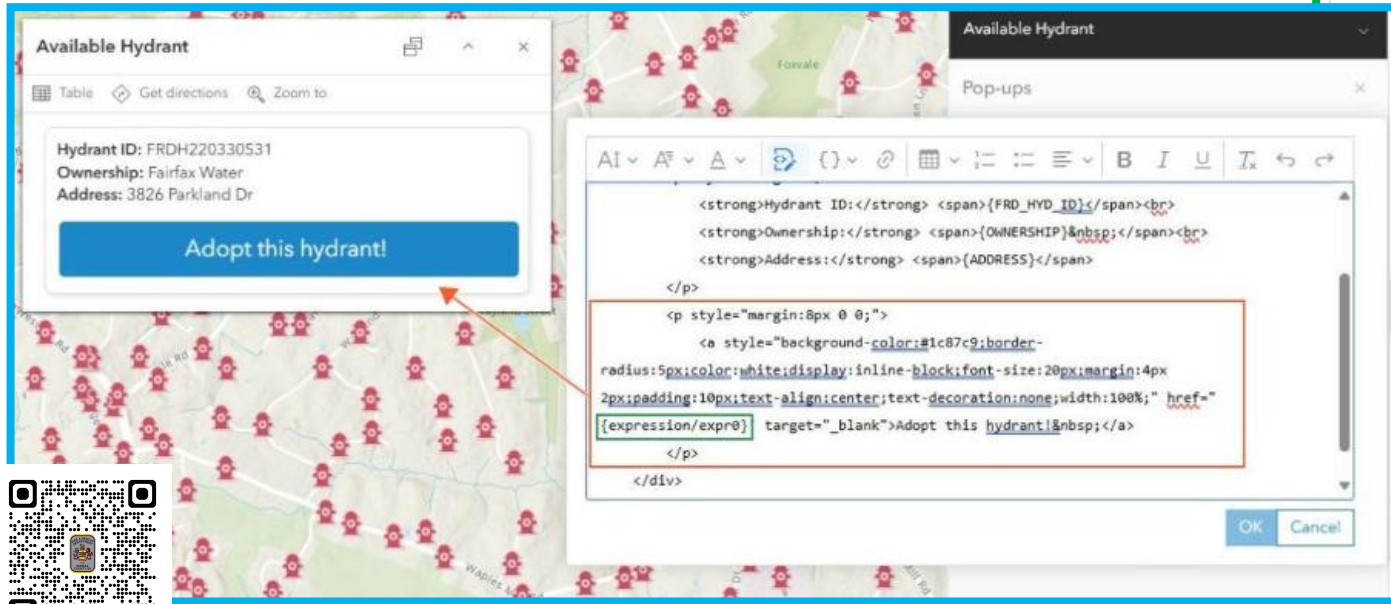
Adopt a Hydrant

Adopt a Hydrant Web Map



- Attribute expressions + custom pop ups = auto-populated survey fields

```
Run
1 // expr0
2 if (!IsEmpty($feature.email)) {
3   return "";
4 }
5 var hydrant_id = $feature.FRD_HYD_ID;
6 var hydrant_address = Replace($feature.ADDRESS, " ", "%20");
7 var ownership = Replace($feature.OWNERSHIP, " ", "%20");
8 var lat = $feature.Y_DD; // Replace with actual field name if different
var lon = $feature.X_DD; // Replace with actual field name if different
var baseUrl = "https://survey123.arcgis.com/share/d1dfaad6b3364337887c66065777b576";
var url = baseUrl + "?open=web" +
  "&field:hydrant_id=" + hydrant_id +
  "&field:hydrant_address=" + hydrant_address
  "&field:ownership=" + ownership +
  "&center=" + lat + "," + lon;
return url;
```



Adopt a Hydrant Experience Builder

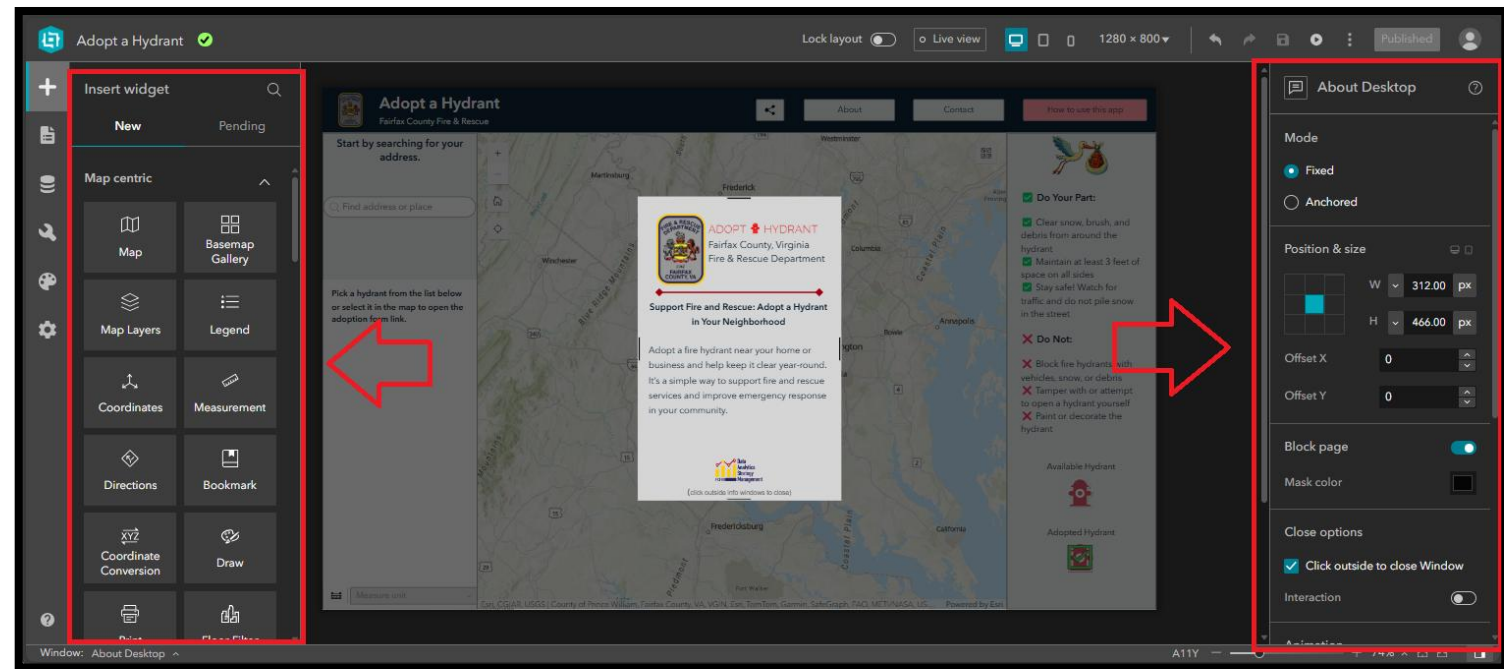


■ Esri's Experience Builder

- web-based application builder that allows you to create custom web apps using ArcGIS Online or ArcGIS Enterprise.

■ Prebuilt widgets

- Search widget used for locating area of interest
- Near Me tool to locate hydrants within 1000 ft

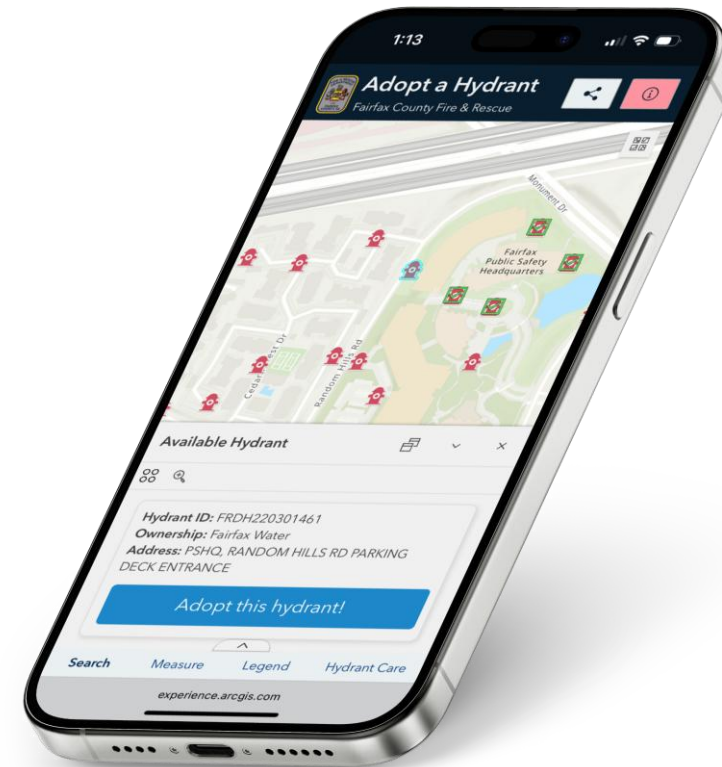
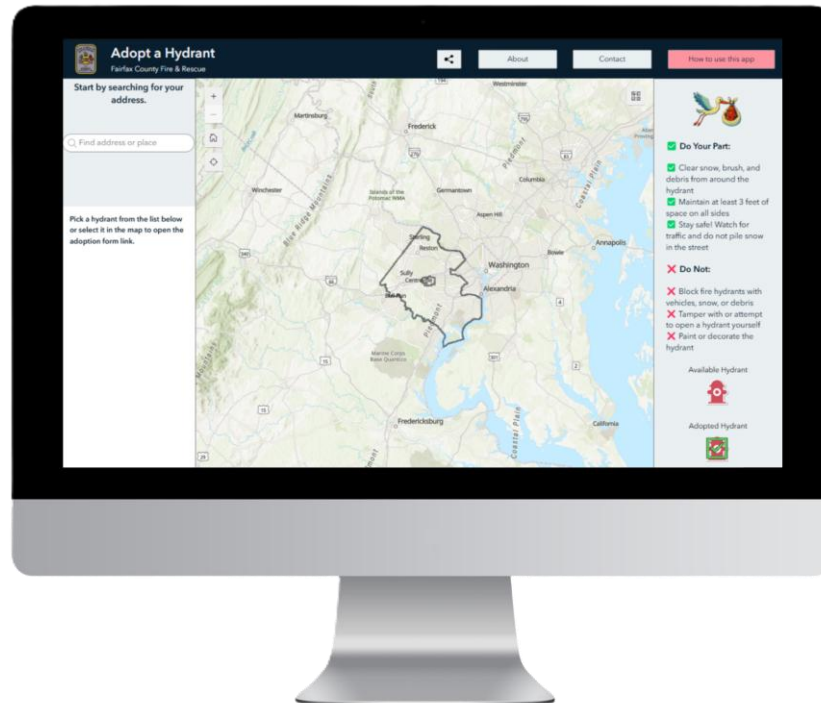


Adopt a Hydrant

Adopt a Hydrant App Features



■ Desktop vs Mobile

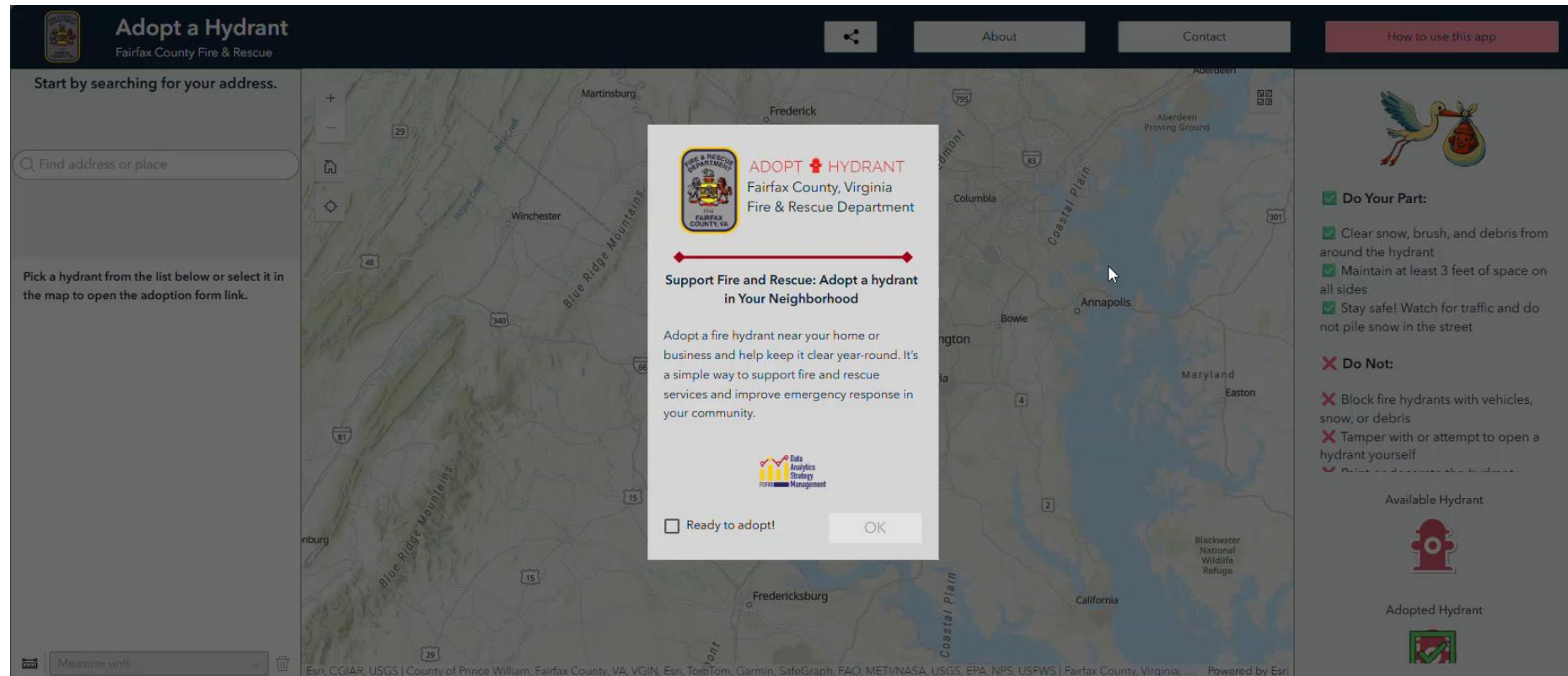


Adopt a Hydrant

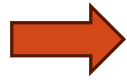
Adopt a Hydrant How to Use



3 step process: Step 1



Search widget



Near Me widget



Adopt a Hydrant

Program Do's & Don'ts



Legend



Adopt a Hydrant How to Use



3 step process: Step 2

Adopt a Hydrant
Fairfax County Fire & Rescue

Start by searching for your address.

Search widget: 12099 GOVERNMENT CENTER PKWY

Search result: 12099 GOVERNMENT CENTER PKWY

Pick a hydrant from the list below or select it in the map to open the adoption form link.

Public Hydrants and Adopted Hydrant Joined View		16
FRDH220301497	141.6 ft	>
FRDH220301436	268.43 ft	>
FRDH220301461	428.21 ft	>
FRDH220301426	469.09 ft	>
FRDH220301389	492.2 ft	>
FRDH220301475	521.07 ft	>
FRDH220301490	638.18 ft	>

Map showing hydrant locations around Monument Dr and Government Center Pkwy.

Program Do's & Don'ts:

- Do Your Part:**
 - Clear snow, brush, and debris from around the hydrant
 - Maintain at least 3 feet of space on all sides
 - Stay safe! Watch for traffic and do not pile snow in the street
- Do Not:**
 - Block fire hydrants with vehicles, snow, or debris
 - Tamper with or attempt to open a hydrant yourself

Legend:

- Available Hydrant (Red hydrant icon)
- Adopted Hydrant (Green hydrant icon)



Adopt a Hydrant

Adopt a Hydrant

How to Use



3 step process: Step 3

The screenshot displays the 'Adopt a Hydrant' web application interface. The header includes the Fairfax County Fire & Rescue logo, the title 'Adopt a Hydrant', and navigation links for 'About', 'Contact', and 'How to use this app'. The main content area is divided into three sections:

- Left Panel:** Contains a search bar with the placeholder 'Find address or place' and instructions: 'Start by searching for your address.' and 'Pick a hydrant from the list below or select it in the map to open the adoption form link.'
- Map:** A map showing various hydrant locations marked with red icons. A pop-up window titled 'Available Hydrant' is displayed over the map, showing details for a specific hydrant: 'Hydrant ID: FRDH220301497', 'Ownership: Fairfax Water', and 'Address: 12099 GOVERNMENT CENTER PKWY'. A blue button labeled 'Adopt this hydrant!' is at the bottom of the pop-up.
- Right Panel:** Contains a list of instructions for users, categorized into 'Do Your Part' and 'Do Not'. It also includes a legend for 'Available Hydrant' (red icon) and 'Adopted Hydrant' (green icon with a checkmark).

At the bottom of the map, there is a footer with copyright information: 'Esri, NASA, NGA, USGS, FEMA, Esri Community Maps Contributors, County of Loudoun, County of Prince William, Fairfax County, VA, VGIN, © OpenStreetMap, Microsoft, Esri, ... Powered by Esri'.



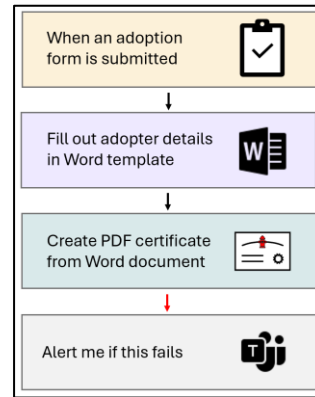
Adopt a Hydrant

Adopt a Hydrant Certificates



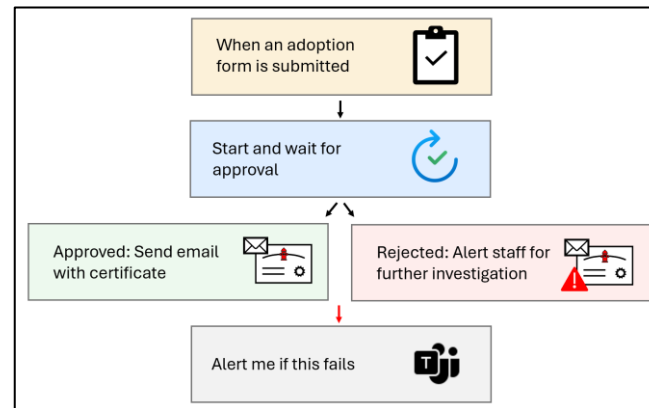
■ Generate Certificate

- Survey123 submission
- Populate Word template
- Save Word doc
- Convert to PDF



■ Staff Review

- Survey123 submission
- Staff review
- Certificate emailed to adopter



Adopt a Hydrant

Adopt a Hydrant Public Outreach

- Multiple Languages
- Website
- Social Media
- News and Radio segments



소화전 지킴이

페어팩스 카운티 소방구조국의 '소화전 지킴이(Adopt a Hydrant)' 프로그램에 참여해 보세요!

여러분 동네의 소화전을 직접 관리하면서 지역 소방서를 지원하고, 지역의 안전을 지키는 데 중요한 역할을 하게 됩니다. 참여 방법은 간단하지만, 이웃과 지역사회에 큰 도움이 되는 뜻깊은 활동입니다.

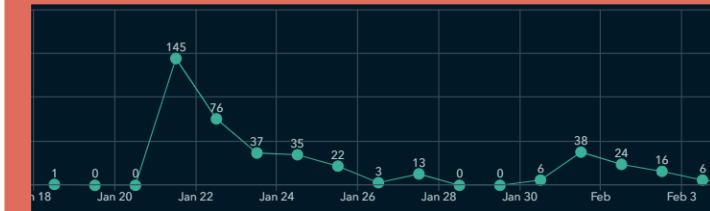
참여하려면 QR 코드를 스캔하시거나 아래 웹사이트를 방문하세요:
<https://www.fairfaxcounty.gov/fire-ems/AdoptAHydrant>



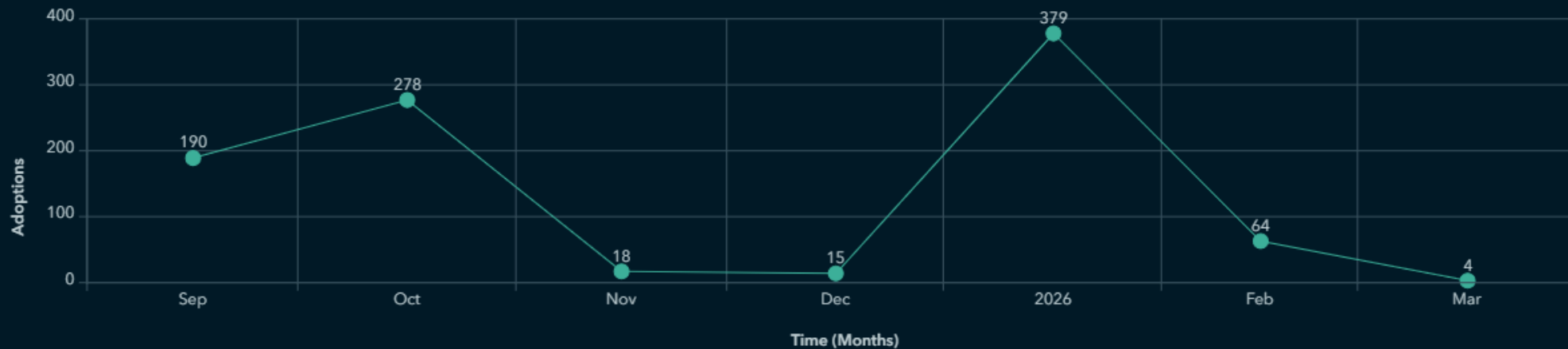
Adopt a Hydrant Progress & Success



On January 22, Governor Spanberger issued [Executive Order 11](#) declaring a state of emergency in advance of the impending winter weather that impacted most of the Commonwealth beginning Saturday, January 24 and lasting through Monday, January 26. A second storm impacted areas mainly south of I-64 lasting beginning Friday, January 30 through Sunday, February 1.



Adoptions Over Time

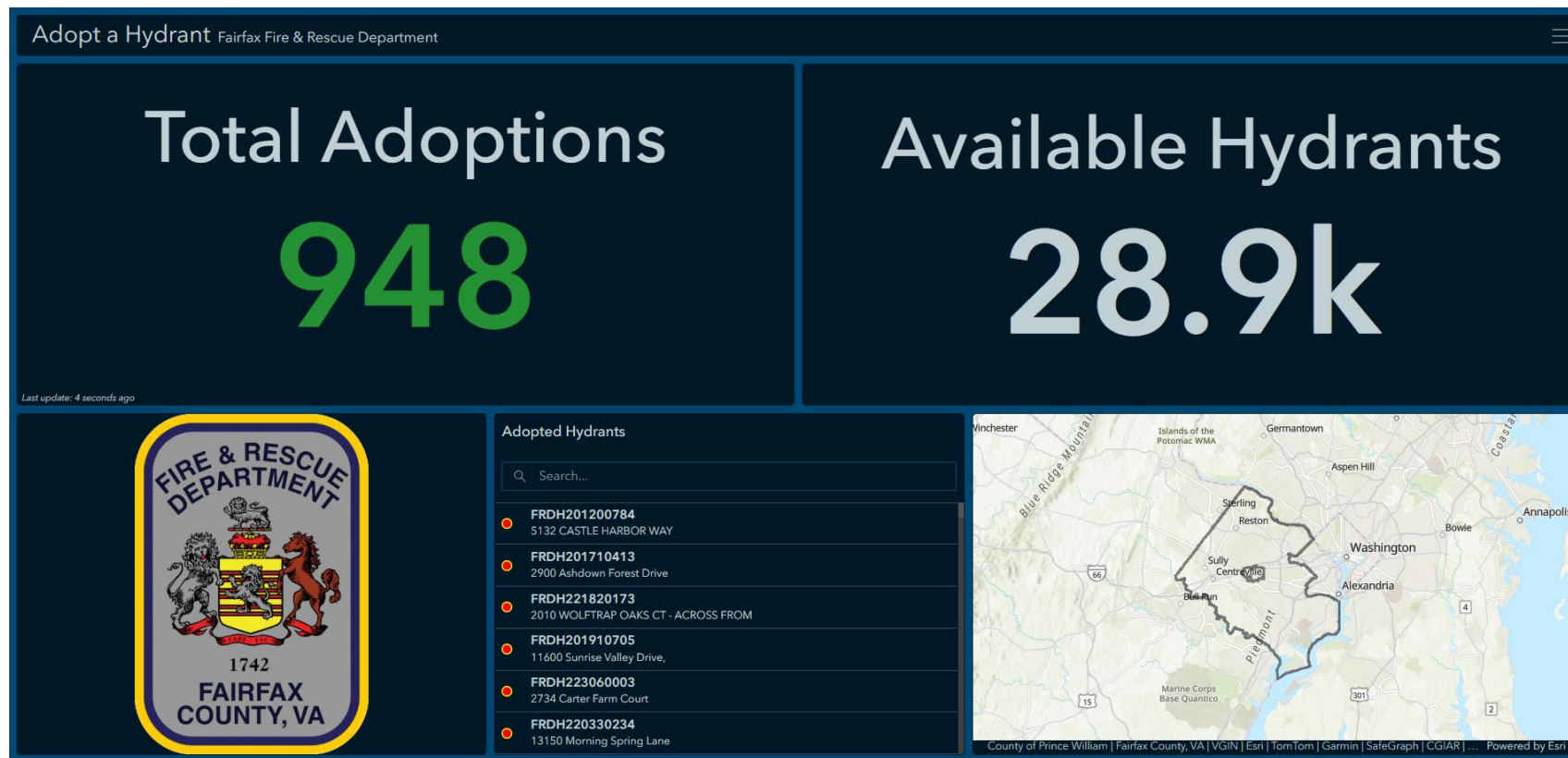


Adopt a Hydrant

Adopt a Hydrant Progress & Success



Internal Dashboard

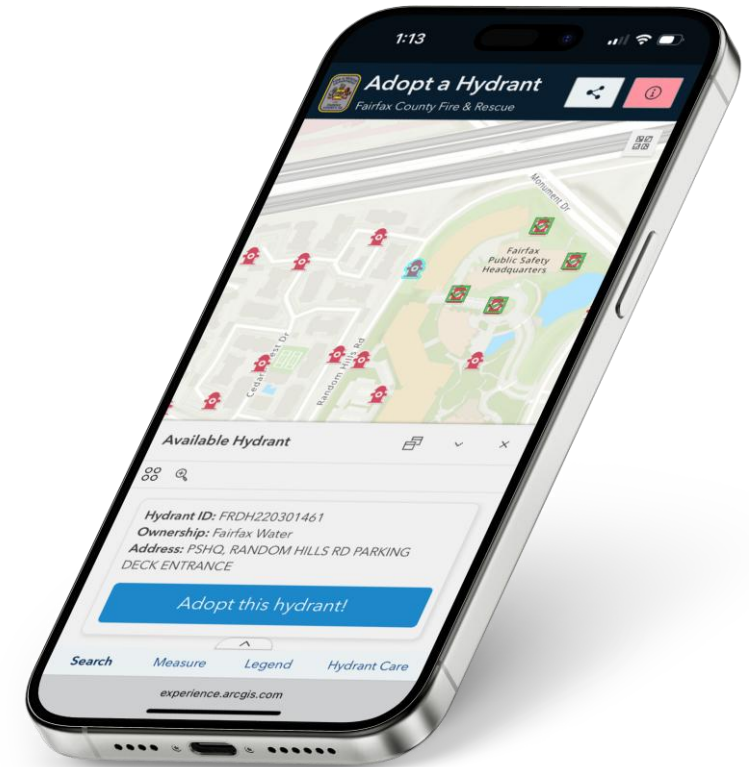


Adopt a Hydrant

Adopt a Hydrant Enhancements



- Next Phase:
 - Name of adopters on hydrant pop-up
 - Can adopters name their hydrant?
 - Link out to a citizen shoveling program (if its created, hinted at it in McKay's Newsletter)
 - Align with other county programs
- Indicator: Total Adopted Hydrants
- Updating UI to match department branding
- Approval Process
 - Daily summary
- Survey123 enhancements
 - Trim function

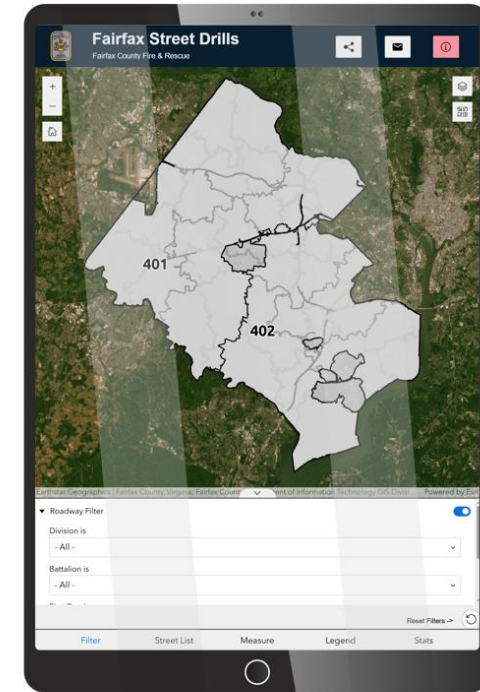
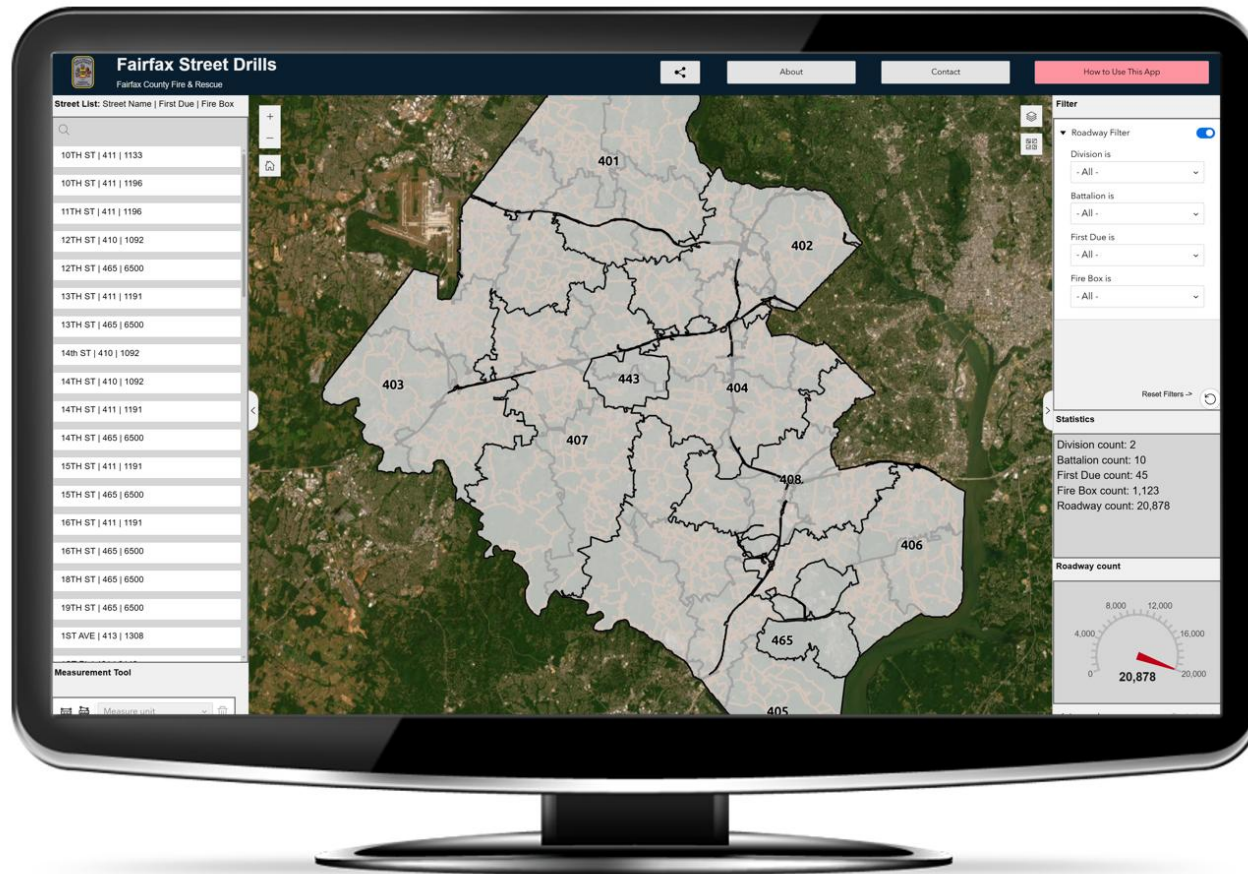


Adopt a Hydrant mobile view



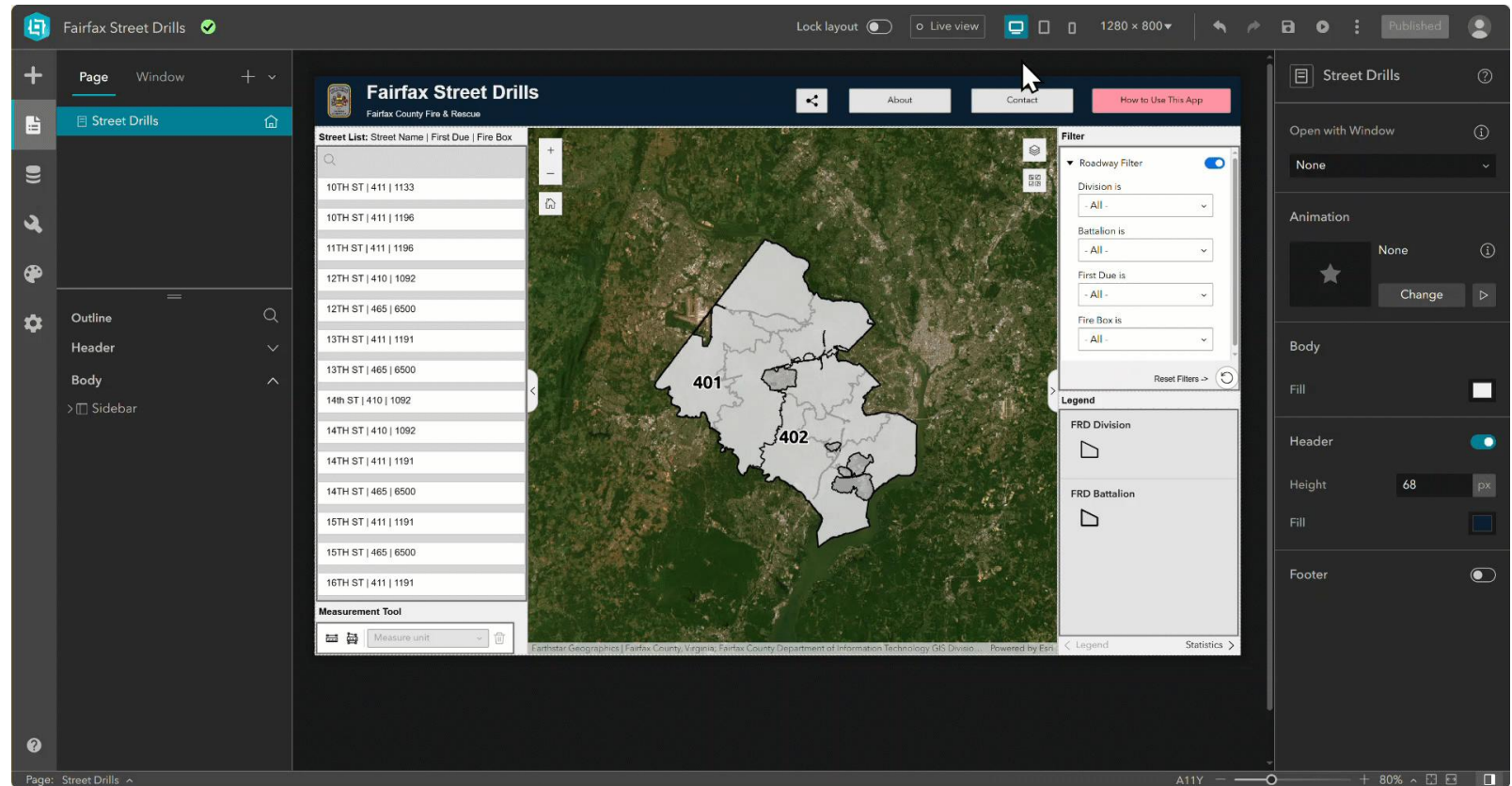
Adopt a Hydrant

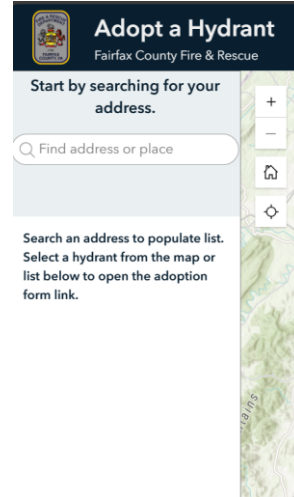
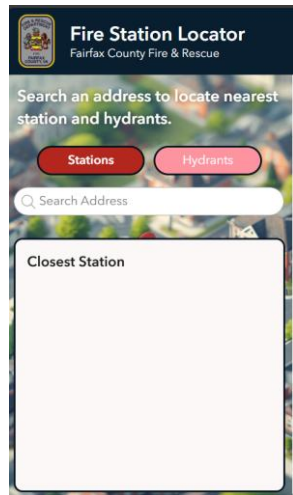
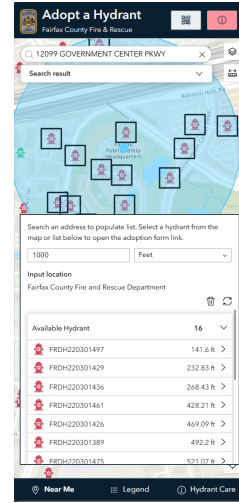
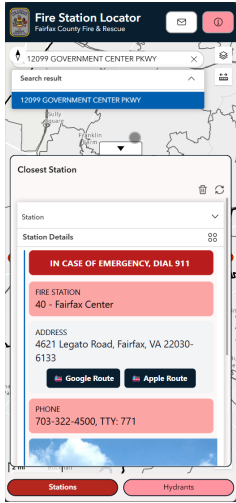
Desktop, Tablet, and Mobile Optimization



Desktop, Tablet, and Mobile Optimization

- Intuitive Workflow
- Vertical Alignment





Desktop, Tablet, and Mobile Optimization

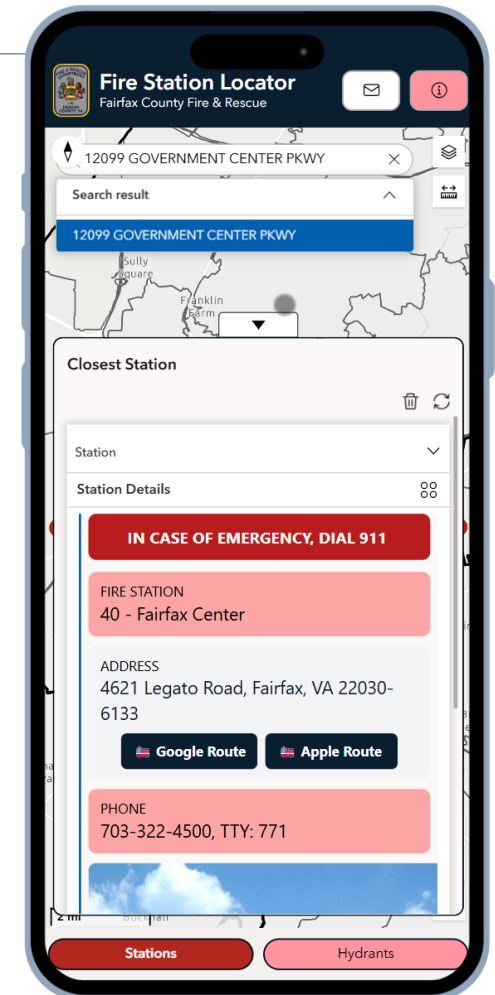
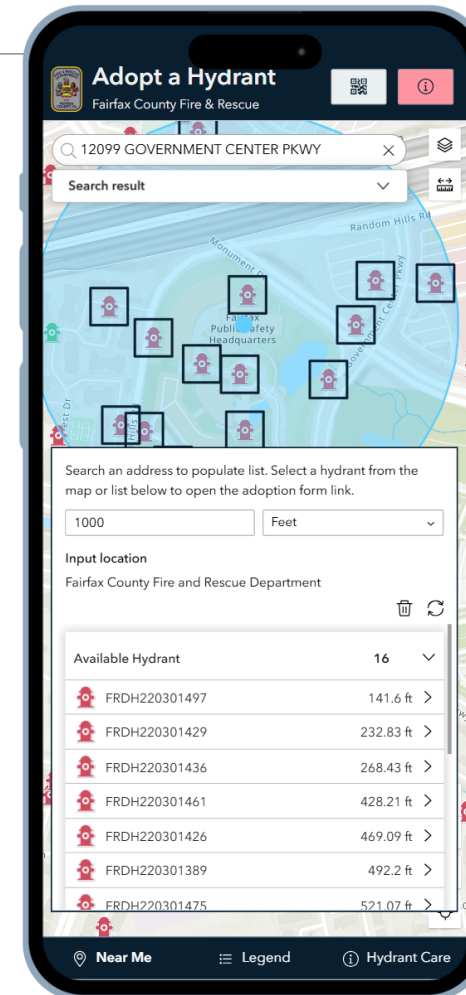
■ Intuitive Workflow

- Minimize clicks/taps to complete common tasks
- Present clear, concise instructions and labels
- Use a simple, linear flow (e.g., Search → Results → Details)
- Keep key widgets and buttons visible and easy to reach
- Ensure the same core workflow works on both desktop and mobile
 - Reduce on-screen clutter; cut out unnecessary widgets

Desktop, Tablet, and Mobile Optimization

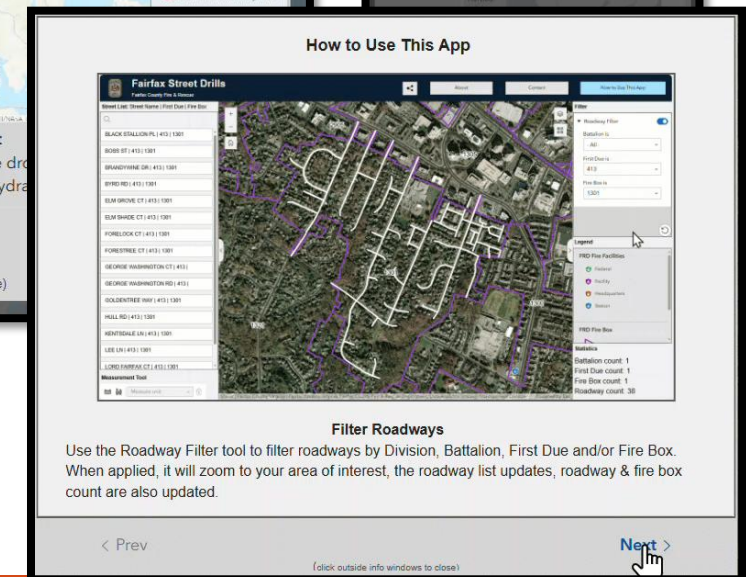
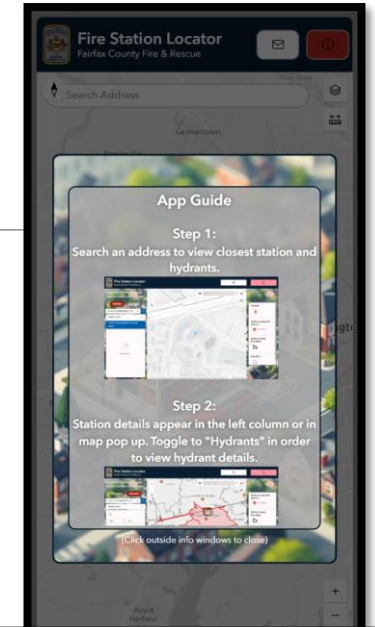
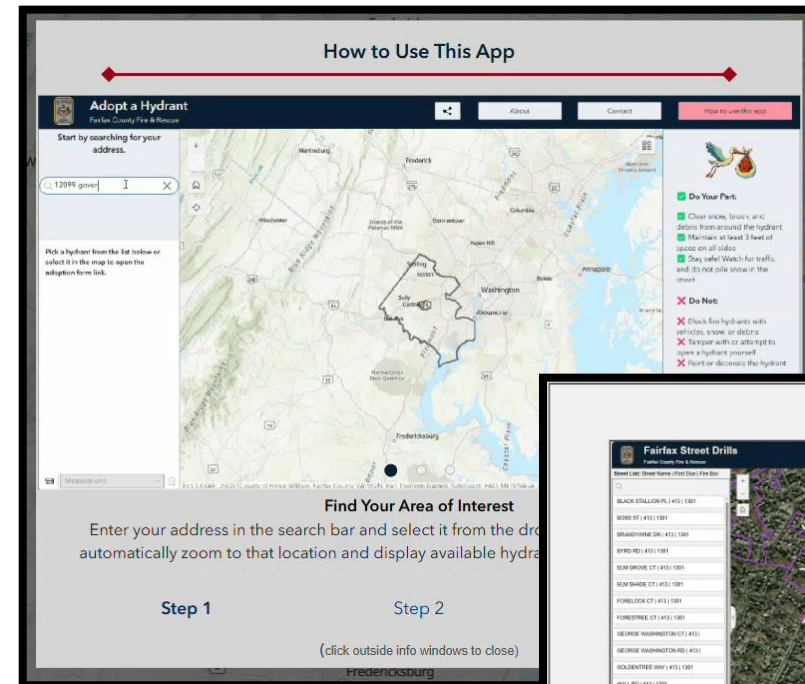
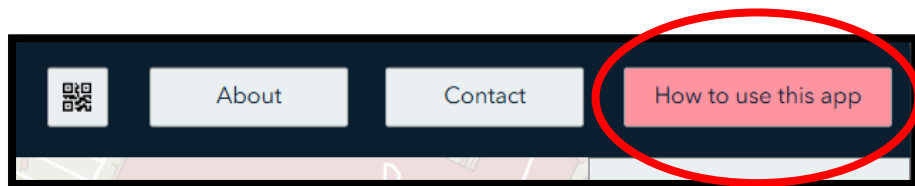
■ Vertical Alignment

- Phones are vertical, screens are small
 - Users scroll up/down, not side-to-side
 - Vertical alignment keeps content in a single, natural reading path
- Thumb-friendly, one-handed use
 - Key buttons and steps should be reachable with a thumb
 - Minimizes zooming, panning, and precision tapping
- Short, guided workflows reduce confusion
 - Clear, step-by-step flow (Search → Results → Details)
 - Fewer clicks and visible options at each stage
- Desktop ≠ mobile
 - Multi-panel desktop layouts don't translate well to small screens
 - Mobile design must prioritize the most important tasks and information



Application Header On-Demand Training

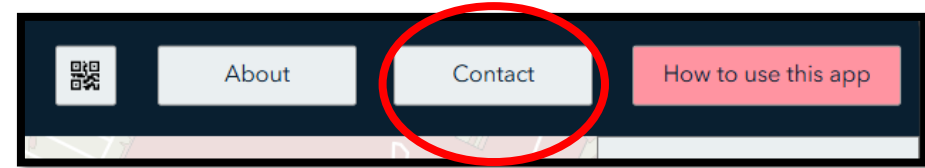
- Helps new users understand the layout, key tools, and workflows without external documentation
- Ensures everyone learns the app the same way, reducing confusion and support questions
- Make sure its easily accessible to users

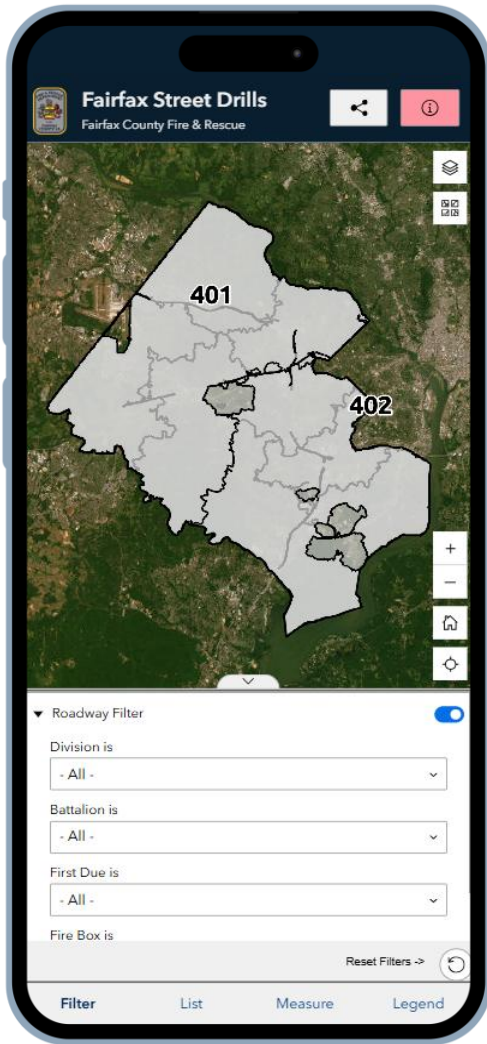


Application Header Contact Us

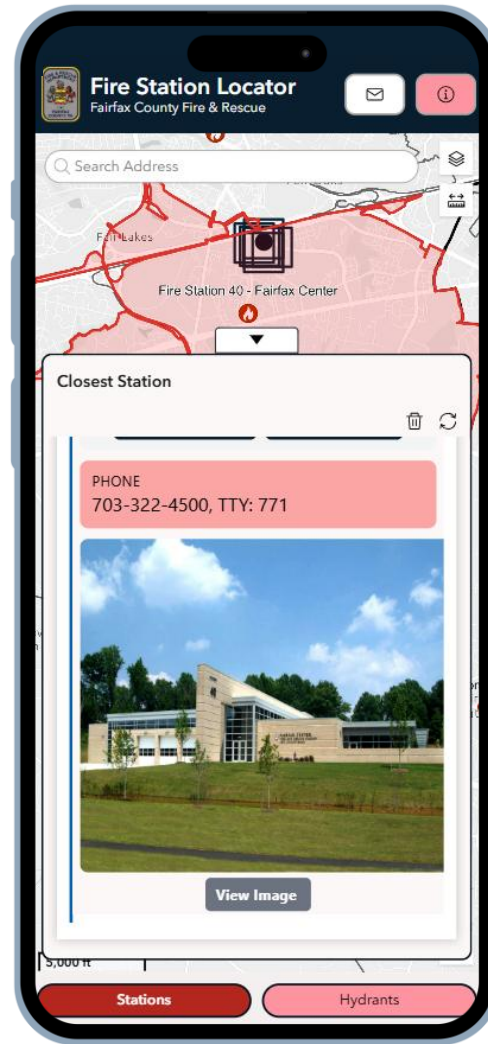
- Hyperlink

- <mailto:fire.GIS@fairfaxcounty.gov?subject=Fire%20Station%20Locator%20Questions%20and%20Suggestions>

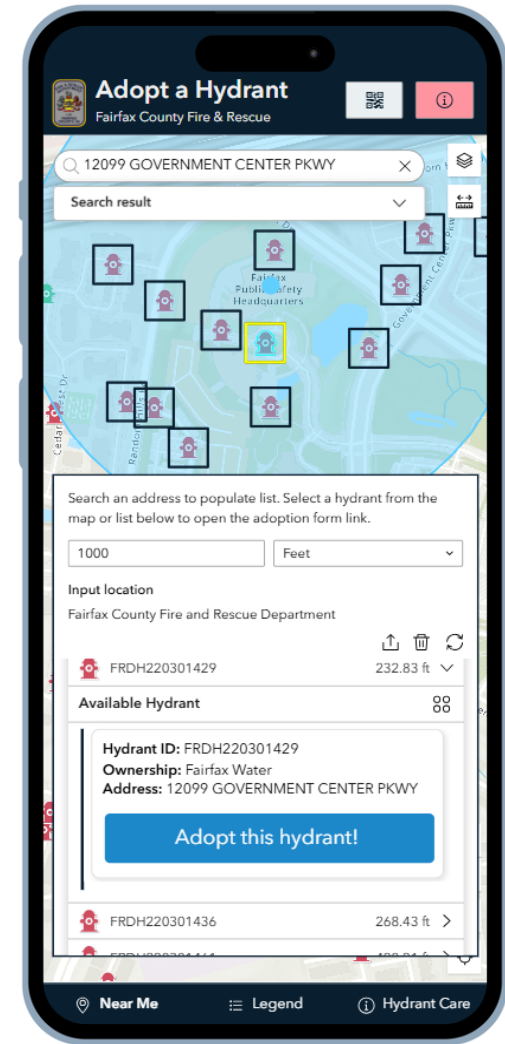




Street Drills



Fire Station Locator



Adopt a Hydrant



Questions?

Contact Info:

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